Supporting Information: ORACLE: A Persistent Chemical State for Reproducible Molecular Workflows

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1 Frozen software and validation boundary

The companion archive defines a single software boundary for the submission. The maintained source repository is `https://github.com/yogibubu/ORACLE-manuscript`; the archive attached to this article freezes the exact release used for the calculations: the integrated ORACLE `0.1.0rc8` source snapshot recorded in the archive manifest. Input files, geometries, scripts, numerical records and checksums are distributed within the same reproducibility boundary:

`ORACLE_data_and_software.zip`

contains the archived source snapshot and the complete validation material. The machine-readable records distributed with the source archive are:

- `validation/topology_corpus_audit.json`;
- `validation/large_ring_systems_audit.json`;

- validation/hbond_fragment_audit.json;
- validation/symmetry_group_audit.json;
- validation/orientation_covariance_probe.json;
- validation/quasisymmetry_threshold_sweep.json;
- validation/nontrivial_symmetrization_audit.json;
- validation/descriptor_invariance_audit.json;
- validation/synthon_atom_class_audit.json;
- validation/switch_cross_platform_validation.json;
- validation/oracle_observable_comparison_saccharin.json;
- validation/saccharin_structure_case.json;
- validation/saccharin_structure_case_frozen_L1.xyzin;
- validation/saccharin_pic_case.json;
- validation/saccharin_lcb25_retrieval.json;
- validation/lcb25_leave_one_out.json;
- validation/lcb25_xtb_differential.json;
- validation/oracle_native_accuracy_ladder.json;
- validation/oracle_rdkit_ring_benchmark.json;
- validation/oracle_rdkit_chemical_cases.json;
- validation/switch_rdkit_benchmark.json;
- validation/oracle_method_basis_sensitivity.json;

- validation/oracle_state_blind_control.json;
- validation/baker_ts_full_manifest.tsv;
- validation/benchmark_uncertainty_summary.json;
- validation/oracle_geometry_perception_performance_report.json;
- validation/oracle_topology_acceleration_benchmark.json;
- validation/oracle_horton_equivalence_audit.json;
- validation/oracle_l0_charge_examples.json;
- validation/oracle_l0_metal_complexes.json;
- validation/gly_ala_gly_fragment_charges.json;
- validation/oracle_gly_ala_gly_fragment_qm_parallel.json;
- validation/hbond_charge_response_parameters.tex;
- validation/perception_scope_audit.json;
- validation/oracle_hexahelicene_lcb25.xyz;
- validation/oracle_hexahelicene_lcb25.xyz.assembly.json;
- validation/oracle_hexahelicene.xyzin.

Each record contains the ORACLE version, a stable release identifier and immutable input fingerprints.

1.1 Reproduction entry point

The remote calculations can be regenerated from a clean checkout of the archived source with the following entry point. The wrapper loads the frozen MATRIX/PySCF environment; the final two commands build the derived controls from the resulting machine-readable records.

```
source ~/.matrix_env.sh
~/matrix/bin/matrix-pyscf validation/run_method_basis_sensitivity.py \
  > validation/oracle_method_basis_sensitivity.json
python3 validation/build_state_blind_control.py
python3 validation/build_state_ablation.py
```

The submission environment is Python 3.13.11, NumPy 2.3.5, and RDKit 2026.03.3. It uses ORACLE release candidate 0.1.0rc8 and the archived MATRIX revision identified in the machine-readable manifest and checksum record. The command is intended for the declared remote environment; local reruns must preserve the same software versions and basis-set data to reproduce numerical values.

The complete perception report follows a versioned ORACLE schema and records the frozen symmetry and topology, the continuous $\sigma/\pi/\pi$–π descriptors, the synthon atom classes, and every primitive definition. Regenerating a report from an existing enriched XYZ therefore verifies the topology and file fingerprints without repeating perception. Independent structures can be processed through a versioned batch manifest; the corresponding regression uses two process workers, verifies distinct outputs, and confirms that their order still matches the manifest.

2 Computational details and software boundary

The integrated ORACLE boundary is organized into versioned sections for the source, chemical identity, topology, rings, synthons, symmetry, fragments, populations, and structure.

Conventional organic systems may enter as SMILES; transition-metal and organometallic systems preferentially enter as an enriched Cartesian record with molecular charge, multiplicity, accepted connectivity, coordination edges, hapticity and metal–metal relationships. The interoperable outer representation follows the QCSchema molecule fields for symbols, geometry, charge, multiplicity and optional connectivity; the versioned enriched-XYZ sections retain the complete perception audit. CXSMILES with dative bonds is accepted only as an exchange representation. If coordinates are absent, a declarative coordination specification records metal centers, charge, multiplicity, ligand representations, donor atoms, denticity, hapticity, bridges, metal–metal bonds and an optional coordination geometry. Template and fragment assembly enumerates Cartesian candidates and coordination stereoisomers, followed by distance, collision, valence, coordination and symmetry audits. Unspecified oxidation or spin states remain separate hypotheses through L0 scoring. QCSchema serialization begins after a Cartesian candidate exists. The executable contract is supplied in the archive as `coordination_input.schema.json`; the companion `coordination_input_examples.json` contains complete coordinate-free ferrocene and cisplatin inputs, and `coordination_input.py` validates identifiers, references, hapticity, denticity, metal–metal links and hypothesis lists before assembly. The workflow internally selects and executes the auxiliary operations needed to obtain atom types and synthons, charges and bond orders, rings and aromaticity, equivalence classes, fragments, and reconstructed or, when requested, optimized geometries. Once a section is complete, its stored definitions become authoritative and are never replaced by an unannounced repetition of perception.

SWITCH is the production chemoinformatics layer. It interprets SMILES, aromaticity, and declared stereochemistry; adds ordinary hydrogens; builds deterministic two- and three-dimensional seeds; provides canonical SMILES and atom ranks; performs substructure/SMARTS and connected maximum-common-subgraph searches; and enumerates unspecified stereoisomers. ORACLE combines the electronic observables supplied by the PySCF interface with continuous perception and assumes ownership of the accepted molecu-

lar state. Fragment-XYZ connectivity, target matching, and overlap assembly consequently rely on ORACLE and SWITCH throughout and retain one authoritative graph.

All calculations reported here passed through SWITCH, with GFN-FF as implemented in xTB 6.7.1 supplying the initial universal force-field geometries and PySCF 2.14.0 the L0 density matrices. The contracts remain backend agnostic, so equivalent force-field and electronic-structure programs can replace either choice. The portable environment requires Python 3.11 or later, NumPy 1.24 or later and SciPy 1.11 or later; Pillow 10 or later supplies PNG rendering. The current archived environment is ORACLE 0.1.0rc8, Python 3.13.11, NumPy 2.3.5, SciPy 1.17.1 and Pillow 12.1.1. C++/OpenMP kernels are optional accelerators with mandatory equivalence to the Python reference.

RDKit 2026.03.3 is retained as an independent conformance, timing, and compact chemical-assignment reference. Production perception, depiction, fragment matching and assembly remain independent of RDKit. The frozen `validation/switch_rdkit_benchmark.json` records 21/21 matches for lexically declared graph semantics, 21/21 preservation tests for canonical isomer identity, 3/3 substructure counts, 2/2 connected MCS sizes and 2/2 stereoisomer counts. SWITCH also realizes opposite Cartesian handedness for `@`/`@@`, the expected E/Z alternatives and a planar square-planar seed. Together with the 58-system aromaticity corpus, these tests provide statistically resolved conformance coverage for every RDKit operation used within the declared workflow.

Table S1: SWITCH conformance against the independent RDKit reference.

Contract	Agreement	Cases
Declared graph semantics	21	21
Canonical isomer identity	21	21
Chiral substructure counts	3	3
Connected MCS atom counts	2	2
Stereoisomer counts	2	2

The same report distinguishes parser scope from seed generation. On the reported arm64 host, median cold SWITCH parsing is 13.1 μs versus 11.6 μs for sanitized RDKit parsing;

the bounded SWITCH cache reduces repeated parsing to 0.084 μs. Median deterministic Cartesian seeding is 1.01 ms versus 0.774 ms for RDKit ETKDG on the small benchmark. The latter comparison is reported only as computational scale because the two initialization algorithms solve different conformational problems. The corresponding seven architecture checks passed on both macOS 26.5.2/arm64 with Python 3.14.3 and Linux 4.18.0/x86-64 with Python 3.11.13. The exact environments and outcomes are stored in `validation/switch_cross_platform_validation.json`; this functional portability check is kept separate from timing portability.

These responsibilities define a simple internal sequence:

1. SWITCH maps names or SMILES to a versioned graph and reproducible seed;
2. the PySCF interface supplies electronic observables for CM5/Mayer analysis;
3. ORACLE maps geometry and optional observables to a persistent state containing continuous evidence and accepted discrete objects; and
4. GFN-FF or another selected backend relaxes a seed while ORACLE preserves the perceived state.

LCB25 access is likewise centralized. The archived resolver downloads the published PCS2, HPCS2 and SE archives once, extracts them into a local cache and writes the `oracle.lcb25.cache.v1` manifest with source URLs, archive hashes, dataset labels and geometry paths. Every use in this work calls the same manifest resolver by dataset and molecule name. Retrieval and validation scripts read the extracted cache, while assembly accepts a resolved `--lcb25-fragment` name. The resolver exposes 331 usable PCS2 geometries and centrally filters the non-geometrical AppleDouble member whose name ends in `.xyz`. Thus record selection is independent of duplicated archive directory names and of the location of the author's checkout. The periodic-table scope of ORACLE and the chemical scope of this accurate library are separate contracts. The geometrical and electronic formulation is defined for the full periodic table, while published LCB25 structures contain H, C,

N, O, F, P, S, and Cl. All LCB25 coverage, retrieval, transfer, and reconstruction statistics therefore refer to that element set. Fixed-geometry L0 examples provide complementary electronic/perception coverage for the other represented elements.

The Avogadro bridge writes a clean first XYZ block and reimports the edited Cartesian coordinates. It then invalidates or regenerates every geometry-dependent section, preventing stale topology or symmetry records from surviving a coordinate change. The regression covers coordinate replacement, schema and fingerprint checks, and deterministic regeneration. The complete adapter inventory, lossless exchange contract and environment-discovery behavior are given in `source/docs/ORACLE_QUICKSTART.md`.

The companion archive is an installable source distribution under the BSD 3-Clause License. It contains package metadata, the exact dependency boundary, a clean-install verifier, examples, and the pinned sources used for all reported results. Third-party packages retain their own licenses.

2.1 Controlled persistence ablation

The persistence component was isolated at the ORACLE handoff. The full route supplied a validated ORACLE state. The ablated route supplied the identical Cartesian source and forced the downstream frontend to reconstruct and serialize the state before compiling the same primitive-coordinate model. Source geometries, compiler options, primitive candidate rules and correctness checks were identical. Table S12 reports the median of five timed runs for each route. The frozen-state route was faster in all eight systems, while the accepted chemical representation and evaluated downstream results were identical in every case. The complete machine-readable record is `validation/oracle_state_ablation.json`; it is generated from the handoff audit by `validation/build_state_ablation.py`.

Table S2: Controlled ablation of persistence at the ORACLE handoff. Times are median milliseconds over five runs.

System	N	No persistence	Persistent state	Speed-up
Water	3	6.523	1.180	5.53
Benzene	12	35.371	8.897	3.98
Thiazole	8	13.028	4.932	2.64
[6]helicene	42	128.510	91.384	1.41
Erythromycin	118	541.302	445.173	1.22
Paclitaxel	113	525.891	422.129	1.25
Rifampicin	117	1226.293	1121.826	1.09
Vancomycin	176	2749.102	2319.238	1.19
Median / median case-wise		327.200	256.757	1.33

3 Complete theoretical formulation

This section brings together the definitions, notation, algorithmic choices, and application equations that support the main article. From a SMILES string or Cartesian XYZ file, ORACLE builds and owns a coherent molecular state containing the imported identity, continuous observables, accepted topology, rings and aromaticity, symmetry and equivalence classes, fragment views, synthons, atom types, charges, and bond orders. The same state stores everything needed for subsequent structural reconstruction or optimization. Graph construction, primitive coordinates, Wilson matrices, and electronic-population acquisition remain internal parts of this process and require no additional user input.

3.1 Persistent molecular-state contract

Before serialization, a validation gate checks atom indexing, geometry–topology consistency, finite-group completeness, primitive definitions, and provenance. Once accepted, the ORACLE sections remain immutable until the molecular identity is deliberately rebuilt.

Table S3: Persistent components of the ORACLE molecular-state contract.

Component	Stored information	Reproducible use
Source	Atomic identities, Cartesian geometry, units, provenance and fingerprints	Unambiguous reconstruction of the imported structure
Symmetry	Group label, operations, atom permutations, residuals, thresholds and equivalence classes	Executable finite-group action and projection
Topology	Weighted graph, canonical cycles, ring systems, incidence, automorphisms and aromatic labels	Reusable graph and ordered ring definitions
Interactions	Hydrogen bonds, pseudocycles, connected components, and ring-centered, interaction-centered and synthon-centered partitions	Auditable fragment resolutions
Descriptors	Coordination, bond-order components, effective atomic numbers, synthons and atom classes	Smooth local comparison
Coordinates	Ordered PICs, values, dimensions, Wilson matrix, rank and fingerprints	Structural targets and coordinate transformations

3.2 Geometrical primitives and analytic Wilson matrix

For Cartesian positions $\mathbf{r}_i$, the distance and valence angle are

$$R_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|, \tag{1}$$

$$\theta_{ijk} = \arccos\left[\frac{(\mathbf{r}_i - \mathbf{r}_j)\cdot(\mathbf{r}_k - \mathbf{r}_j)}{R_{ij}R_{kj}}\right]. \tag{2}$$

Dihedrals use an atan2 form built from the normals $\mathbf{n}_1 = (\mathbf{r}_j - \mathbf{r}_i) \times (\mathbf{r}_k - \mathbf{r}_j)$ and $\mathbf{n}_2 = (\mathbf{r}_k - \mathbf{r}_j) \times (\mathbf{r}_l - \mathbf{r}_k)$. Out-of-plane coordinates use a signed sine form; near-linear angles use two components in a deterministic orthonormal frame perpendicular to the axis.

ORACLE constructs ordered stretches, bends, linear bends, torsions and out-of-plane terms from the frozen covalent and interaction records. Each primitive stores its atom tuple,

type, dimension, value and provenance. The analytic Wilson matrix is

$$B_{\alpha k} = \frac{\partial q_\alpha}{\partial x_k}. \tag{3}$$

Its dimensions, numerical rank and geometry/definition fingerprints are serialized. For active targets S,

$$\Delta \mathbf{x} = \mathbf{B}_S^T \left(\mathbf{B}_S \mathbf{B}_S^T + \lambda \mathbf{I} \right)^{-1} \Delta \mathbf{q}_S. \tag{4}$$

When no Hessian metric is available, the deformation hierarchy is torsion < angle < stretch, with weights 1, 100 and 1000. Disconnected covalent components are closed by perceived interaction pseudo-bonds and, only if necessary, a deterministic nearest-pair minimum spanning tree.

3.3 Continuous coordination and unique bond-order contract

Local connectivity begins with the continuous coordination number (CNA),

$$\mathrm{CNA}_i = \sum_{j \neq i} g_{ij}, \qquad g_{ij} = \frac{1}{2} \left[1 + \mathrm{erf}\big(\alpha_{\mathrm{CNA}} (R_{ij}^0 - R_{ij}) \big) \right], \tag{5}$$

where $R_{ij}^0 = R^{\mathrm{cov}}(Z_i) + R^{\mathrm{cov}}(Z_j)$. The coordination-dependent radius $\widetilde{R}_i^{\mathrm{cov}} = \mathcal{H}(Z_i, \mathrm{CNA}_i)$ uses a C^1 cubic Hermite interpolation of Pyykkö radii. The internal topology weight

$$w_{ij}^{\mathrm{conn}} = \exp\left[\frac{\widetilde{R}_i^{\mathrm{cov}} + \widetilde{R}_j^{\mathrm{cov}} - R_{ij}}{\lambda_{\mathrm{conn}}} \right] \tag{6}$$

provides the internal evidence from which topology is accepted. Downstream consumers nevertheless see one unambiguous serialized chemical bond order.

ORACLE likewise exposes one array of atomic charges and one array of bond orders. Complete Charge Model 5 (CM5) atomic charges and Mayer bond orders select the paired quantum-mechanical (QM) level for the whole structure;[S1,S2] if either array is incomplete,

the state consistently uses the paired electronegativity/Pauling level. This all-or-nothing choice prevents mixed-provenance observables. The geometry-only estimate is

$$\mathrm{BO}_{ij} = \exp\left[\frac{R_i^{\mathrm{cov,MT}} + R_j^{\mathrm{cov,MT}} - R_{ij}}{0.30\ \text{Å}}\right]. \tag{7}$$

Define the one-sided C^2 excess

$$\mathcal{E}_\delta(x) = \begin{cases} 0, & x \le 0, \\ \delta(6t^3 - 8t^4 + 3t^5),\ t = x/\delta, & 0 < x < \delta, \\ x, & x \ge \delta, \end{cases} \qquad \delta = 0.10. \tag{8}$$

With $P_1 = \mathcal{E}_\delta(\mathrm{BO} - 1)$ and $P_2 = \mathcal{E}_\delta(\mathrm{BO} - 2)$,

$$\mathrm{BO}^{(\sigma)} = \mathrm{BO} - P_1, \tag{9}$$

$$\mathrm{BO}^{(\pi)} = P_1 - P_2, \tag{10}$$

$$\mathrm{BO}^{(\pi\pi)} = P_2. \tag{11}$$

The three non-negative components sum identically to the unique bond order. Atomic indices are $I_i^{(a)} = \sum_{j \in \mathcal{N}(i)} \mathrm{BO}_{ij}^{(a)}$. The geometry-only electronegativity charge is

$$q_i^{\mathrm{EN}} = \sum_{j \in \mathcal{N}(i)} \frac{\chi_j - \chi_i}{(n_i + n_j)\max(\mathrm{BO}_{ij}, \mathrm{BO}_{\min})}. \tag{12}$$

3.4 Synthon classes and GAFF2 translation

Native atom types emerge from element-preserving complete-link agglomeration of the seven atomic-synthon components. A merge is accepted only while the full range of every component remains below its declared threshold, thereby preventing threshold chaining. Each serialized class records its atoms, centroid, maximum componentwise spread, and threshold vector.

The optional GAFF2 translation consumes the accepted adjacency, aromatic atoms and continuous bond-order decomposition. The current conservative map assigns `ca` to aromatic carbon; `c`, `c1`, `c2` and `c3` to carbonyl, *sp*, sp^2 and sp^3 carbon; `n`, `na`, `nb`, `n2`, `n3` and `n4` to amide, aromatic and ordinary nitrogen environments; and `o`, `oh` and `os` to carbonyl, hydroxyl and ether oxygen. Hydrogen follows its accepted parent as `ha`, `hc`, `hn`, `ho`, `hp` or `hs`. Phosphorus uses `p2`, `p3`, `p4` or `p5` according to accepted coordination and multiple-bond evidence. Sulfur uses `s`, `s2`, `s4`, `s6`, `sh` or `ss`. Ambiguous or unsupported environments produce an explicit unsupported record, so the diagnostic translation cannot silently alter the native synthon class or any perception result.

3.5 L0 populations, fragment synthesis and hydrogen bonds

The default L0 population request is a fixed-geometry, gas-phase PBE0/def2-TZVP single point. ORACLE calls the PySCF application-programming interface directly, passing Cartesian coordinates, charge, and multiplicity in memory and selecting restricted or unrestricted Kohn–Sham theory consistently with the electronic state. After tight SCF convergence, the atomic-orbital overlap matrix, one-particle density, molecular-orbital coefficients, and occupations pass directly to population-analysis interface, which derives CM5 charges and Mayer bond orders from that same state. No external input, separate front end, or text-output parser intervenes.

The def2-TZVP basis is used on every atom. For Rb–Rn only the heavy centers receive the corresponding native def2 ECP; lighter atoms in a mixed system remain all-electron. The population analysis restores the ECP core contribution to the effective nuclear charge before applying CM5. The hydrogen-bond training complexes, ferrocene and cisplatin were all evaluated at their supplied fixed geometries; no geometry optimization was performed. Every reported job reached the requested SCF convergence, and the complete request, geometry, electronic-state specification and provenance are retained in the machine-readable record.

PBE0 is used here to provide stable structural and population descriptions across the

periodic table. Its nonempirical PBE/exact-exchange construction avoids tuning to a particular energetic data set, while CM5/Mayer observables were selected because their trends vary comparatively little under reasonable changes of method and basis. Competitive thermochemical energies are outside the role assigned to this level.

Let fragment f contain an owned core $\mathcal{C}_f$ and a graph-dilated halo $\mathcal{H}_f$. The union $\mathcal{C}_f \cup \mathcal{H}_f$ is evaluated quantum mechanically, but only core observations reconstruct the molecular population. With positive weights w_f,

$$W_i^{\mathrm{C}} = \sum_{f:i\in\mathcal{C}_f} w_f, \tag{13}$$

$$\bar{q}_i = (W_i^{\mathrm{C}})^{-1} \sum_{f:i\in\mathcal{C}_f} w_f q_{if}, \tag{14}$$

$$\sigma_{i,\mathrm{ov}}^2 = (W_i^{\mathrm{all}})^{-1} \sum_{f:i\in\mathcal{C}_f\cup\mathcal{H}_f} w_f q_{if}^2 - \left[(W_i^{\mathrm{all}})^{-1} \sum_{f:i\in\mathcal{C}_f\cup\mathcal{H}_f} w_f q_{if} \right]^2 . \tag{15}$$

The default planner gives every atom exactly one core owner and dilates each core by two bond-graph shells. The outer halo boundary alone is capped. Every accepted covalent pair must occur in at least one retained fragment, all fragments use the same electronic level, and the overlap graph must be connected. Halo observations enter $\sigma_{i,\mathrm{ov}}$ and related boundary diagnostics but never the assembled charge. The molecular charge is restored by the weighted minimum-norm projection

$$q_i = \bar{q}_i - \frac{\sum_j \bar{q}_j - Q_{\mathrm{mol}}}{W_i^{\mathrm{C}} \sum_j (W_j^{\mathrm{C}})^{-1}}. \tag{16}$$

For each Mayer pair, observations are ranked by whether two, one or zero endpoints belong to the fragment core; only the highest available rank is averaged. Core and total overlap counts, $\sigma_{i,\mathrm{ov}}/\sqrt{n_i^{\mathrm{all}}}$, cap mappings and the maximum charge projection are serialized.

Local charge groups are constructed inside each accepted covalent component only after core/halo synthesis is complete. Their integer targets sum to the molecular charge: every group in a neutral component has zero target, whereas one explicit group carries the residual

charge of a charged component. With $\mathbf{A}$ the matrix of group and atom-type constraints, the physical charges solve

$$\min_{\mathbf{q}} \frac{1}{2}(\mathbf{q}-\bar{\mathbf{q}})^T\mathbf{W}(\mathbf{q}-\bar{\mathbf{q}}) \quad \text{subject to} \quad \mathbf{A}\mathbf{q}=\mathbf{b}. \tag{17}$$

The resulting closed-form weighted minimum-displacement solution is accompanied by rank and residual audits. Neutral groups remove the monopole term from their far field and can pass directly to fast-multipole consumers. Because they are formed after population synthesis, these electrostatic groups remain conceptually and operationally distinct from the overlapping QM fragments.

Let $\mathbf{q}^{\text{obs}}$ be the CM5 vector calculated at the stored reference geometry $\mathbf{R}_{\text{ref}}$. The same perception kernel used at runtime determines the complete reference contact set and strengths $s_c^{\text{ref}} = s_c(\mathbf{R}_{\text{ref}})$. The intrinsic, hydrogen-bond-free vector is

$$\mathbf{q}^0 = \mathbf{q}^{\text{obs}} - \sum_c s_c^{\text{ref}} \Delta\mathbf{q}_c, \tag{18}$$

and the runtime vector is

$$\mathbf{q}(\mathbf{R}) = \mathbf{q}^0 + \sum_c s_c(\mathbf{R})\Delta\mathbf{q}_c = \mathbf{q}^{\text{obs}} + \sum_c \left[s_c(\mathbf{R}) - s_c^{\text{ref}}\right]\Delta\mathbf{q}_c. \tag{19}$$

Thus the model describes charge fluctuations about an observed CM5 reference, while retaining a well-defined no-hydrogen-bond limit. Equations 18 and 19 are an algebraic inverse pair: the former must be applied exactly once during state construction, and the latter at every subsequent geometry. Their round-trip residual at $\mathbf{R}_{\text{ref}}$ is recorded in the serialized ORACLE state.

The locality assumption applies to the fluctuation. The CM5 reference already represents the electronic distribution at $\mathbf{R}_{\text{ref}}$; a response term is introduced only for a recognized change expected to produce a significant additional polarization or charge transfer. The current

production library covers hydrogen bonds. The persistent ORACLE state can also detect coordination-number changes, bond-order rearrangements, ion pairs, metal coordination and ring closure/opening. These are candidate response events, not automatically transferable hydrogen-bond rules. In the absence of an event-specific calibration, the runtime emits an out-of-domain diagnostic and preserves the reference model.

For each contact,

$$\Delta\mathbf{q}_c = \Delta\mathbf{q}_c^{D,\mathrm{pol}} + \Delta\mathbf{q}_c^{A,\mathrm{pol}} + \Delta\mathbf{q}_c^{\mathrm{CT}}, \tag{20}$$

subject to

$$\sum_{i\in D} \Delta q_i^{D,\mathrm{pol}} = 0, \qquad \sum_{i\in A} \Delta q_i^{A,\mathrm{pol}} = 0, \qquad \sum_{i} \Delta q_i^{\mathrm{CT}} = 0. \tag{21}$$

The selected X–H pair defines the donor polarization support, expanded to N–C=O for formamide. On the acceptor side, the support comprises A and its attached hydrogens for water or ammonia, C=O for formaldehyde, and N–C=O for formamide. Within these chemically local supports, the fitted response is the minimum-L^2 projection of the dimer-minus-isolated-monomer CM5 vector onto the stated neutrality constraints. The residual change in the donor fragment total defines interfragment charge transfer; by convention, a negative stored value denotes electron transfer from acceptor to donor.

The CM5 input is classified explicitly. The default `auto` policy maps a supplied CM5 vector to an `environment_polarized` observation: it was calculated on the supplied reference structure and must be depolarized with Equation 18. The alternative `intrinsic` flag is accepted only for a charge vector already obtained without the contacts represented by the response library. Mixing these two contracts would respectively double count or omit the reference response and is therefore recorded as a provenance error. If CM5 data are unavailable, `auto` classifies the electronegativity estimate as intrinsic because it contains no explicit environment response.

The distance factor is the ratio of generalized noncovalent Pauling orders,

$$s_{\mathrm{HB}}(r) = \frac{B_{\mathrm{vdW}}(r)}{B_{\mathrm{vdW}}(r_{\mathrm{ref}})}, \qquad B_{\mathrm{vdW}}(r) = \exp\left[-\frac{r}{R_{\mathrm{vdW},H} + R_{\mathrm{vdW},A}}\right]. \tag{22}$$

The resident UFF van der Waals radii already used by perception determine the range, so this expression introduces no additional fitted parameter. The correction follows the lifetime of the ORACLE hydrogen-bond contact, while the L0 Mayer H$\cdots$A value at $r_{\mathrm{ref}} = 1.85$ Å serves as an electronic quality-control observable; the contact geometry remains the interpolation coordinate.

For water, let

$$\nu_D = \sum_{h=1}^{2} s_h, \qquad \nu_A = \sum_{a=1}^{2} s_a, \qquad 0 \le \nu_D, \nu_A \le 2 \tag{23}$$

be the continuous donated- and accepted-contact occupancies of one molecule. The water parameters are inherited from the model of Ref. S3, not refitted from new clusters. Its integer counts are replaced by ν_D and ν_A, the isolated reference by the symmetry-projected L0 PBE0/def2-TZVP observation, and the saturated reference by the TIP3P-FB vector:

$$\mathbf{q}^{\mathrm{CM5}} = (-0.674180563,\, 0.337090282,\, 0.337090282)e,$$
$$\mathbf{q}^{\mathrm{FB}} = (-0.84844,\, 0.42422,\, 0.42422)e.$$

If $\Delta q_H = q_H^{\mathrm{FB}} - q_H^{\mathrm{CM5}}$, the symmetric three-site specialization is

$$\begin{aligned} \Delta q_{H_k}^{\mathrm{pol}} &= \frac{\Delta q_H}{2} s_k + \frac{\Delta q_H}{4} \nu_A, \\ \Delta q_O^{\mathrm{pol}} &= -\sum_{k=1}^{2} \Delta q_{H_k}^{\mathrm{pol}}, \end{aligned} \tag{24}$$

with the two acceptor capacities represented symmetrically on oxygen. Each donated contact adds the published $+\beta$ fragment charge and each accepted contact adds $-\beta$, where $\beta = 0.10e$;

atomwise assignment follows Equations 19 and 20 of Ref. S3. Thus $Q_{\mathrm{CT}} = \beta(\nu_D - \nu_A)$. For the central water of the symmetric pentamer, $\nu_D = \nu_A = 2$: charge transfer cancels, Equation 24 gives $\mathbf{q} = \mathbf{q}^{\mathrm{FB}}$ exactly, and molecular charge remains zero. This endpoint follows directly from the published parameter constraint and is independent of a new pentamer fit.

Each s_h and s_a is an analytic distance function and the response is linear in the continuous occupancies, so both first and second Cartesian derivatives are analytic. The response can therefore be evaluated directly, without an induced-charge iteration. At an ellipsoidal nonperiodic boundary, a smooth exposure function supplies the missing occupancy $2-\nu_D$ or $2-\nu_A$ toward the same TIP3P-FB endpoint. Boundary completion is polarization only: it creates no fictitious partner and no charge transfer. The simulation layer thereby preserves the condensed-phase limit without periodic images or a global polarization solve.

ORACLE compiles donor/acceptor sites and their synthon response rules once, before a trajectory begins. During propagation, the shared chemistry runtime retains covalent adjacency, component membership, and a Verlet-style H$\cdots$A pair list, rebuilding the list only after an atom has moved by more than half the declared skin. Downstream applications can thus update contacts and occupancies locally during Monte Carlo, molecular dynamics, or optimization without re-entering ORACLE. Explicit site lists extend the same kernel beyond the default N/O/P/S chemistry, for example to the acidic C–H donor and weak chlorine acceptors of chloroform.

For a non-water–water contact, no separate global model is introduced. The runtime selects the donor-synthon/acceptor-synthon response fitted from the corresponding rigid dimer and adds its polarization and charge-transfer vectors with the analytic contact strength. The published water specialization and the dimer-derived generalization consequently share one recognition and derivative contract without sharing parameters.

3.6 Deterministic geometry and population route

Cartesian XYZ coordinates are preserved when supplied, and an explicitly requested optimization proceeds through L0→L1→PL1. For a SMILES input, the default construction route is SWITCH→GFN-FF→L0→L1→PL1. SWITCH first interprets the complete graph and declared stereochemistry and then builds a deterministic Cartesian seed. From the accepted bond orders, ORACLE infers ordinary missing valences, places hydrogens by maximizing their angular separation from existing bonds, and records every addition. Explicit SMILES hydrogen counts take precedence in cases such as aromatic `[nH]`; phosphorus, sulfur, metals, and ambiguous charged environments remain fail-closed unless that count is explicit.

GFN-FF supplies the first relaxation because its parameterization spans the periodic table and offers a more accurate geometry description than UFF for the biochemical, organometallic, and materials systems considered here.[S4,S5] Each subsequent level receives the final updated approximate Hessian from the preceding optimization as its initial Hessian; this reuse of optimizer state requires no explicit frequency calculation. Whenever compatible LCB25/LCB26 fragments are available, their validated local geometries supersede the universal starting model. Large targets are divided into connected, overlapping fragments until atoms, covalent pairs, geometries, and populations are all covered. The definitions and calibrations of L1 and PL1 used in the examples otherwise remain unchanged.

3.7 Electronic domains, angular strain and synthons

$$N_{\pi,i} = \sum_{j\in\mathcal{N}(i)} \left[\mathrm{BO}_{ij}^{(\pi)} + \mathrm{BO}_{ij}^{(\pi\pi)}\right], \tag{25}$$

$$N_{\mathrm{res},i} = n_i^{\mathrm{val}} - \mathrm{CNA}_i - N_{\pi,i}, \tag{26}$$

$$N_{\mathrm{OS},i} = \mathrm{mod}(\mathrm{round}(N_{\mathrm{res},i}), 2), \tag{27}$$

$$N_{\mathrm{LP},i} = \max\left(0, \frac{N_{\mathrm{res},i} - N_{\mathrm{OS},i}}{2}\right), \tag{28}$$

$$N_{\mathrm{ED},i} = \mathrm{CNA}_i + N_{\mathrm{LP},i} + N_{\mathrm{OS},i}. \tag{29}$$

For integer domain number N, $\bar{\theta}(N) = \Theta(N)/\binom{N}{2}$; noninteger values use cubic Hermite interpolation.

Table S4: Reference angular means used for strain evaluation.

N	2	3	4	5	6	8	12
$\bar{\theta}$ (deg)	180.0	120.0	109.47	120.0	90.0	90.0	63.43

The compactness and radial-dispersion terms are

$$C_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j\in\mathcal{N}(i)} \frac{\widetilde{R}_i^{\mathrm{cov}} + \widetilde{R}_j^{\mathrm{cov}}}{R_{ij} + \widetilde{R}_i^{\mathrm{cov}} + \widetilde{R}_j^{\mathrm{cov}}}, \tag{30}$$

$$D_i = \frac{1}{|\mathcal{N}(i)|\bar{R}_i} \sum_{j\in\mathcal{N}(i)} |R_{ij} - \bar{R}_i|. \tag{31}$$

With $D_i^{\mathrm{dom}} = N_{\mathrm{ED},i}$, the sole production strain is

$$S_i = \left[\frac{1}{\binom{D_i^{\mathrm{dom}}}{2}} \sum_{j<k} \left(\cos\theta_{ijk} - \cos\theta_i^{\mathrm{ref}}\right)^2\right]^{1/2}. \tag{32}$$

The unique atomic synthon is

$$\mathbf{s}_i = (q_i, C_i, D_i, S_i, I_i^{(\sigma)}, I_i^{(\pi)}, I_i^{(\pi\pi)}). \tag{33}$$

Environmental tuning uses

$$\mathbf{x}_i^{(\mathrm{ET})} = \left(\operatorname{erf}\frac{q_i}{\sigma_{\mathrm{ET}}}, \operatorname{erf}\frac{C_i}{\sigma_{\mathrm{ET}}}, \operatorname{erf}\frac{D_i}{\sigma_{\mathrm{ET}}}, \operatorname{erf}\frac{S_i}{\sigma_{\mathrm{ET}}}\right), \quad \mathrm{ET}_i = \frac{\|\mathbf{x}_i^{(\mathrm{ET})}\|_2}{1 + \|\mathbf{x}_i^{(\mathrm{ET})}\|_2}, \tag{34}$$

and $Z_i^{\mathrm{eff}} = Z_i - \frac{1}{2} + \mathrm{ET}_i$. The atom metric is

$$d_{ij}^{(\mathrm{atom})} = \left[(Z_i^{\mathrm{eff}} - Z_j^{\mathrm{eff}})^2 + \|\mathbf{s}_i - \mathbf{s}_j\|_2^2\right]^{1/2}. \tag{35}$$

Bond, angle and dihedral synthons compose the same atomic state:

$$\mathcal{S}_{ij}^{(\mathrm{bond})} = (\mathbf{s}_i, \mathbf{s}_j, \mathrm{BO}_{ij}), \tag{36}$$

$$\mathcal{S}_{ijk}^{(\mathrm{angle})} = (\mathcal{S}_{ij}^{(\mathrm{bond})}, \mathcal{S}_{jk}^{(\mathrm{bond})}, \cos\theta_{ijk}), \tag{37}$$

$$\mathcal{S}_{ijkl}^{(\mathrm{dihedral})} = (\mathcal{S}_{ijk}^{(\mathrm{angle})}, \mathcal{S}_{jkl}^{(\mathrm{angle})}, \cos\phi_{ijkl}, \cos^2\phi_{ijkl}). \tag{38}$$

This recursive construction preserves the provenance of every higher-order descriptor: expanding any tuple recovers the same atomic synthons, bond orders, and geometrical variables. A ring synthon follows the ordered bond and angle synthons of its canonical cycle and adds the ring geometry. A hydrogen-bond synthon similarly joins the donor, hydrogen, and acceptor synthons with the contact and interaction scores, distances, angle, and interaction class.

3.8 Gaussian molecular and ring similarity

Atomic embeddings $\mathbf{y}_i = (Z_i^{\mathrm{eff}}, \mathbf{s}_i)$ are summarized by a regularized Gaussian $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. For molecules A and B, with $\boldsymbol{\Sigma} = \frac{1}{2}(\boldsymbol{\Sigma}_A + \boldsymbol{\Sigma}_B)$,

$$D_B = \frac{1}{8}\Delta\boldsymbol{\mu}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\Delta\boldsymbol{\mu} + \frac{1}{2}\left[\ln\det\boldsymbol{\Sigma} - \frac{1}{2}(\ln\det\boldsymbol{\Sigma}_A + \ln\det\boldsymbol{\Sigma}_B)\right], \tag{39}$$

and $\mathrm{Sim}(A,B) = \exp(-D_B)$. Rings use size, planarity, normalized radial position, fused degree, nearest-centroid distance and ring radius in the same construction. The general score is

$$\mathrm{Sim}_{\mathrm{comb}} = (1 - w_{\mathrm{ring}})\,\mathrm{Sim} + w_{\mathrm{ring}}\,\mathrm{Sim}_{\mathrm{ring}}, \qquad w_{\mathrm{ring}} = 0.25, \tag{40}$$

with the ring term disabled when both molecules are acyclic. The coverage-directed kernels based on the published LCB25 local-geometry library[S6] are defined in the dedicated section below.

3.9 Hydrogen bonds and pseudocycles

For donor D, bonded hydrogen H and acceptor A,

$$h_{HA} = \frac{1}{2}\left[1 + \mathrm{erf}\big(\alpha_{\mathrm{HB}}(R_H^{\mathrm{vdW}} + R_A^{\mathrm{vdW}} - R_{HA})\big)\right], \tag{41}$$

and

$$\mathrm{HBO}_{HA} = \exp\left[\frac{R_H^{\mathrm{vdW}} + R_A^{\mathrm{vdW}} - R_{HA}}{\lambda_{\mathrm{HB}}}\right]\frac{1 + \cos(\pi - \theta_{DHA})}{2}. \tag{42}$$

$\alpha_{\mathrm{HB}} = 1/\sigma_{\mathrm{HB}}$ is a parameter conversion, not a second model. λ_{HB} is interaction-class provenance. Accepted intramolecular contacts close a pseudocycle through the minimal covalent path; intermolecular contacts remain interaction edges between distinct covalent components. General perception covers compatible N, O, P and S sites. The resident charge-response table contains the complete O–H/N–H/P–H/S–H by O/N/C=O/P/S product. Each of its 20 cells is the mean of 16 rigid parent or substituted dimers; bootstrap confidence intervals and leave-one-environment-out shifts record statistical stability.

3.10 Canonical cycles, ring systems and automorphisms

Ring perception begins from the covalent non-metal graph. Deterministic shortest-path trees generate Horton candidates, and any chorded perimeter is split recursively. Ordering

the candidates by length and canonical atom sequence allows binary Gaussian elimination to select a minimum-length independent basis of cardinality $|E|-|V|+c$. For $C = (v_1, \ldots, v_m)$,

$$\mathrm{can}(C) = \min\left(\{\rho_k(C)\}_{k=0}^{m-1} \cup \{\rho_k(C^{\mathrm{rev}})\}_{k=0}^{m-1}\right). \tag{43}$$

The serialized result includes the basis, its completeness flag, atom–ring and bond–ring incidence, ring systems, and fused connectivity. No maximum ring size is imposed in production.

Ring automorphisms must extend to the complete labeled molecular graph. Vertex labels combine element, synthon signature, external attachments and ring incidence; edge labels combine bond class and ring incidence. Thus an unsubstituted five-cycle has ten maps, oxolane retains only two, and constitutionally different substitutions can reduce the orbit to identity. Aromaticity is separate: absent an electronic-state override, a ring must have at least five atoms, be planar and give every member a compatible non-metal valence environment.

3.11 Executable point-group symmetry

After center-of-mass translation, the inertia tensor is

$$\mathbf{I} = \sum_i m_i \left[(\mathbf{r}_i \cdot \mathbf{r}_i)\mathbf{1} - \mathbf{r}_i \mathbf{r}_i^{\mathsf{T}}\right]. \tag{44}$$

Candidate operations require element-consistent mappings

$$\|\mathbf{R}_g \mathbf{r}_i - \mathbf{r}_{\pi_g(i)}\| < \tau, \qquad Z_{\pi_g(i)} = Z_i. \tag{45}$$

The inertia frame generates candidates but is not a symmetry assignment. Symmetric tops anchor the degenerate plane to a deterministic off-axis atom. Spherical tops recover point-set isometries from noncoplanar triples pruned by element, radius and Gram matrix. Accepted matrices are conjugated back to the input frame.

Each operation is

$$g = (\mathbf{R}_g, \pi_g), \qquad \mathbf{R}_g^{\mathsf{T}}\mathbf{R}_g = \mathbf{1}, \qquad \det \mathbf{R}_g = \pm 1. \tag{46}$$

The set must contain $h \circ g = (\mathbf{R}_h\mathbf{R}_g, \pi_h \circ \pi_g)$, identity and inverses. Residuals and margins are

$$\epsilon_g = \max_i \|\mathbf{R}_g\mathbf{r}_i - \mathbf{r}_{\pi_g(i)}\|, \qquad m_g = \tau - \epsilon_g. \tag{47}$$

The default absolute Cartesian threshold and relative inertia tolerance are 10^{-3} in their respective units. Quasi-symmetry requires a stable group/order plateau, positive margins and an appropriate projection displacement. The smallest plateau threshold is preferred unless $\epsilon_{\max}/\tau > 0.8$. For a complete group,

$$\mathbf{r}_i^{\mathrm{sym}} = \frac{1}{|G|}\sum_{g\in G} \mathbf{R}_g^{\mathsf{T}}\mathbf{r}_{\pi_g(i)}. \tag{48}$$

Maximum and RMS displacements are reported. Projection is not performed for C_1, an incomplete group or an input residual below 5×10^{-6} Å. Polar vectors transform as $\mathbf{v} \mapsto \mathbf{R}_g\mathbf{v}$ and axial vectors as $\mathbf{a} \mapsto \det(\mathbf{R}_g)\mathbf{R}_g\mathbf{a}$. Primitive equivalence is the orbit $\mathcal{O}(p) = \{g \cdot p \mid g \in G\}$.

3.12 Accuracy-ladder structural targets

L0 denotes the PBE0/def2-TZVP baseline defined above; L1, PL2, L2 and L3 retain the published valence-level definitions used by the archived structural examples and the underlying high-level and templating developments.[S7–S10] Core–valence (CV) is a transversal layer; BL1 adds conjugation residuals to L1 and PL1 extends BL1 to selected pair classes. For the

a posteriori CV model,

$$w_{AB} = \exp\left[-\left(\frac{R_{AB} - R^0_{AB}}{1.3R^0_{AB}}\right)^2\right], \quad R^0_{AB} = R^{\mathrm{cov}}_A + R^{\mathrm{cov}}_B, \tag{49}$$

$$\Delta R^{\mathrm{CV}}_{AB} = w_{AB}(\alpha_A + \alpha_B), \tag{50}$$

$$\alpha_Z = R^{\mathrm{cov}}_Z(\beta_{0,p} + \beta_{1,p} n_v). \tag{51}$$

The period-line pairs are

$$(\beta_{0,2}, \beta_{1,2}) = (-4.1889094413, 0.5212770805), \tag{52}$$

$$(\beta_{0,3}, \beta_{1,3}) = (-7.2988006002, 0.8169634536). \tag{53}$$

The result is in milliangstrom before conversion; H has zero amplitude and no extrapolation is made beyond H, B–F and Al–Cl.

The BL1 local form is

$$\Delta R_X = A_X \exp\left[-\left(\frac{R_X - R^{\mathrm{ref}}_X}{\sigma_X}\right)^2\right] g_X, \tag{54}$$

where for original C=C and C=S classes $g_X = 1$, $A_X = -\Delta R^{\mathrm{CV}}_X$, and

$$\sigma_X = \frac{1}{1.5} \min\left(|R_{\mathrm{double}} - R_{\mathrm{single}}|, |R_{\mathrm{triple}} - R_{\mathrm{double}}|\right). \tag{55}$$

For audit only, the arithmetic midpoint of the two saccharin polar-sulfur residuals is

$$\bar{\Delta R}^{\mathrm{diag}}_{\mathrm{polar\ S}} = -0.003310 \text{ Å}, \qquad (Z_i, Z_j) \in \{(8,16), (7,16)\}. \tag{56}$$

It is not a calibrated model parameter and is not transferred. The separate legacy high-level

C–H correction is $\Delta R_{\mathrm{C-H}} = -0.000870$ Å. The calibrated PL1 class is

$$\Delta R_{\mathrm{H\cdots O}} = 0.030 \exp\left[-\left(\frac{R_{\mathrm{H\cdots O}} - 2.15}{0.50}\right)^2\right] \frac{1 + \cos(\pi - \theta_{\mathrm{OHO}})}{2} \quad \text{Å}. \tag{57}$$

The general perception layer covers nitrogen, oxygen, phosphorus and sulfur; the displayed correction is restricted to the calibrated O–H$\cdots$O(H) class.

3.13 Scalability and notation

Neighbor lists and local descriptors scale with the accepted local neighborhoods. Primitive internal-coordinate construction is likewise region-local: stretches follow accepted edges, whereas bends and torsions follow bounded graph neighborhoods. Deterministic cycle candidates combined with binary independence avoid enumerating all simple cycles. The large-ring timings exclude three-dimensional embedding and serve as diagnostics, while synthetic 16–512-vertex cycles isolate topological size from chemical complexity. Biological polycycles add chemically realistic tests of connectivity, incidence, primitive generation, and the full $3N - 6$ rank.

Table S5: Core notation.

Symbol	Meaning
g_{ij}, CNA_i	Smooth coordination weight and continuous coordination number
w_{ij}^{conn}	Internal topology weight, not a bond order
BO_{ij}	Unique bond order at the selected availability level
$I_i^{(\sigma,\pi,\pi\pi)}$	Incident bond-order component sums
C_i, D_i, S_i	Compactness, radial dispersion and angular strain
$\mathbf{s}_i$, Z_i^{eff}	Atomic synthon and effective atomic number
h_{HA}, HBO_{HA}	Hydrogen-bond contact and interaction order
$g = (\mathbf{R}_g, \pi_g)$	Cartesian operation and atom permutation
τ, ϵ_g, m_g	Symmetry threshold, residual and margin
$q_\alpha, \mathbf{B}$	Primitive coordinate and Wilson matrix

4 Topology and aromaticity corpus

The topology corpus was selected before evaluating the publication candidate and includes isolated, fused, spiro, cage and metallocene graphs. Expected cycle sizes and aromatic-atom counts are explicit chemical assertions, not values inferred by accepting the previous program output. A pass requires the expected bond count, elementary cycle-basis rank and sizes, aromatic-atom count and a single connected fragment.

Table S6: Topology and aromaticity audit; every row passed all five checks.

Case	Bonds	Rings	Ring sizes	Aromatic atoms
Benzene	12	1	6	6
Pyrrole	10	1	5	5
Azulene	19	2	5, 7	10
Pyrene	29	4	6, 6, 6, 6	16
Spiro system	32	2	6, 6	0
Cubane	20	5	4, 4, 4, 4, 4	0
Ferrocene, eclipsed	30	2	5, 5	10
Ferrocene, staggered	30	2	5, 5	10
Pro–Gly	24	1	5	0
Ribose	19	1	6	0
Thujone	28	2	3, 5	0

4.1 Serialized ring and ring-system automorphisms

The topology writer adds a versioned `AUTOMORPHISMS` block after ring perception. Separate orbits are stored for each elementary ring and each connected ring system. Regression tests recover all ten maps of an unsubstituted five-cycle, reduce oxolane to the two maps consistent with the heteroatom label, and reduce a five-cycle bearing constitutionally different carbon substituents to the identity because no nontrivial ring map extends to the complete labelled graph. Additional fused, spiro, bridged and cage tests verify that every serialized permutation preserves vertex labels, bond labels and adjacency. These records are topology outputs for downstream coordinate phasing; no puckering coordinate is constructed in this work.

4.2 Large-ring and biological-polycycle audit

The public interface accepts either SMILES or XYZ input. For this timing audit, the four large molecular inputs are archived XYZ files so that three-dimensional embedding is not included in the perception time; companion PubChem isomeric simplified molecular-input line-entry system (SMILES) strings and Secure Hash Algorithm 256-bit (SHA-256) hashes preserve identity and provenance. No bonds, cycles or atom selections are supplied. The program constructs the complete state from element labels and Cartesian coordinates. The validation checks atom count, cycle-size multiset, equality of basis count and cyclomatic rank, and the full $3N - 6$ primitive B rank.

Table S7: Large-system cycle and primitive audit.

Case	Atoms	Rings	PICs	B rank	Ring sizes
Erythromycin	118	3	831	348	6, 6, 14
Paclitaxel	113	7	778	333	4, 6, 6, 6, 6, 6, 8
Rifampicin	117	5	762	345	5, 6, 6, 6, 24
Vancomycin	176	10	1173	522	6, 6, 6, 6, 6, 6, 6, 12, 16, 16

The JavaScript Object Notation (JSON) record additionally stores every ring as its exact canonical one-based atom sequence. For example, the erythromycin macrolide is {3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 16, 18}; the rifampicin ansa ring is {2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 20, 21, 23, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34}; and the two 16-membered vancomycin cycles are {17, 18, 19, 38, 39, 41, 42, 74, 73, 62, 63, 64, 65, 66, 67, 68} and {19, 20, 21, 22, 23, 24, 25, 26, 29, 30, 31, 33, 34, 35, 37, 38}. These are the ordered definitions consumed by downstream incidence and coordinate constructors, not post hoc structural annotations.

Synthetic isolated cycles containing 16, 32, 64, 128, 256 and 512 carbon vertices test the absence of a hidden size cutoff independently of chemical perception. Every graph gives one Horton candidate, one selected basis cycle, and a complete rank-one cycle space. Wall times, revision identifiers and all per-case checks are stored in `validation/large_ring_systems_audit.json`; timings are diagnostic and are not used as pass criteria.

The audit exposes why continuous evidence and a discrete aromatic label must remain

distinct. Used alone, a nonzero delocalization proxy promotes strained saturated rings in cubane and thujone, whereas a fused junction in pyrene falls below the corresponding binary threshold. The released rule therefore combines the continuous descriptor with planarity and a conservative non-metal valence environment. Rings smaller than five atoms require an explicit electronic-state override, and metal–ring coordination contacts do not increase the non-metal degree used to establish ferrocene eligibility.

The resulting binary label answers the graph-level question required by downstream typing and coordinate construction, whereas the continuous descriptor preserves a graded measure of the underlying evidence. This separation is useful because the geometry-based harmonic oscillator model of aromaticity (HOMA) and nucleus-independent chemical shifts (NICS) probe different observables and can disagree with each other as well as with a perception rule. Electronic states outside the validated domain are handled through an explicit override.[S11,S12]

5 Performance and external benchmarks

5.1 Compiled topology path and Python-reference equivalence

The numerically dominant topology path is implemented in the optional C++ extension

`matrix_chem._chem_native`.

It evaluates pair geometry, continuous coordination, Hermite-interpolated radii, admissible connectivity, cyclic-core restriction, Horton candidates, cycle-space rank, and binary independence while the Python global interpreter lock is released. Runtime selection remains explicit: if the extension is unavailable or the user requests the reference backend, the pure-Python implementation remains operative and produces the same serialized molecular-state schema.

The median speedup is 5.74, with a range of 2.24–9.93, making compiled execution a

Table S8: Compiled topology acceleration relative to the Python reference.

Case	Atoms	Python (μs)	Compiled (μs)	Speedup
Water	3	90.7	40.5	2.24
Benzene	12	524.9	156.6	3.35
Naphthalene	18	962.4	223.3	4.31
Testosterone	49	4520.6	630.3	7.17
Paclitaxel	113	15075.3	1517.8	9.93
Vancomycin	176	36474.9	3987.8	9.15

material part of the accessible system-size range. Correctness is protected independently of timing: all six paired results are identical, the cyclic-core result agrees with complete Horton construction for 4991/4991 graphs, and the wider graph/ring audit passes 346/346 cases. The extension reports compiler, architecture, OpenMP, and build metadata at runtime. The Python fallback provides the architecture-independent reference against which future macOS, Linux, x86-64, arm64, and accelerator builds are required to be tested. Raw records are in `validation/oracle_topology_acceleration_benchmark.json` and `validation/oracle_horton_equivalence_audit.json`.

5.2 ORACLE performance contract

Both the compiled and Python-reference paths use conservative distance preperception, restrict continuous refinement to admissible candidates, cache radii, bond-order quantities and cycle rank, and use a symmetric 3×3 eigensolver for ring planes. The compiled path additionally removes Python object traffic from the dominant graph/cycle kernels. The five dense continuous matrices are retained as a compact local-pair state and materialized together only on first access.

Table S9: Single-environment timing summary.

Scope	ORACLE	RDKit
Geometry median (ms)	0.0509	0.0570
Geometry p95 (ms)	0.0841	0.1156
Fixed-graph median (ms)	0.0101	0.0195

On this environment, ORACLE is 1.12-fold faster than RDKit at the median, 1.37-fold faster at p95, and 1.94-fold faster for the fixed-graph task. Explicit materialization of CONNECTIVITY, BO, BO_SIGMA, BO_PI, and BO_PI_PI requires 0.1116 ms at the median. Graph, ring-size, cycle-rank, and benchmark checks pass in 346/346 cases, and the cyclic-core basis agrees with complete Horton construction for 4991/4991 molecules. Publication values and the independent local rerun are stored in `validation/oracle_geometry_perception_performance_report.json` and `validation/oracle_rdkit_ring_benchmark.json`, respectively.

5.3 ORACLE–RDKit comparison

The external ring benchmark contains 346 structures: the 331 CCSD(T)/complete-basis-set plus core–valence geometries of the published LCB25 local-geometry library at its PCS2 reference level,[S6] the 11 curated topology cases, and erythromycin, paclitaxel, rifampicin, and vancomycin. Python 3.13.11, NumPy 2.4.2, and RDKit 2026.03.3 were used on arm64 macOS. Each reported time is the median of 15 runs after one untimed warm-up. The complete environment, per-case results, and source paths are frozen in `validation/oracle_rdkit_ring_benchmark.json`.

The timing script retains all 15 repetitions for every molecule and method. A deterministic paired hierarchical bootstrap resamples both molecular cases and within-case runs. The resulting 95% intervals are 1.09–1.15 for the geometry-first RDKit/ORACLE ratio and 1.90–2.03 for the fixed-graph ratio. They capture run-level and molecular-corpus variation on this environment; portability across hardware is examined separately, and no cross-platform speed claim is inferred. The procedure and output are frozen in `validation/summarize_benchmark_uncertainty.py` and `validation/benchmark_uncertainty_summary.json`.

Two timing scopes provide matched workloads. In the fixed-graph scope, both methods receive the same ORACLE nonmetal covalent edge set. ORACLE constructs its canonical cycle basis, whereas RDKit constructs its smallest set of smallest rings (SSSR) with

`Chem.GetSSSR`. In the geometry-first scope, ORACLE constructs continuous and discrete connectivity before its cycle basis. RDKit starts from the same elements and Cartesian coordinates and calls `rdDetermineBonds.DetermineConnectivity` with `useHueckel=False`, charge 0, covalent-radius factor 1.3, and `useVdw=True`, followed by `Chem.GetSSSR`. Metal vertices are excluded from both cycle spaces. Cycle-space completeness is verified by rank over the two-element field GF(2) against $|E|-|V|+c$; matching ring counts remain a separate descriptive quantity.

Table S10: ORACLE–RDKit ring benchmark summary.

Metric	ORACLE	RDKit
Curated bond/ring references	11/11	11/11
Fixed-graph complete cycle space	346/346	346/346
Fixed-graph median time (ms)	0.0101	0.0198
Geometry-first median time (ms)	0.0511	0.0569
Geometry-first p95 time (ms)	0.0846	0.1163

The fixed-graph cycle-size multisets agree in 346/346 cases. From geometry, the complete edge sets and resulting cyclomatic ranks agree in 338/346 cases. At fixed graph ORACLE is 1.96-fold faster. From geometry ORACLE is 1.11-fold faster at the median and 1.38-fold faster at p95. This path performs conservative preperception and continuous refinement for admissible candidates, and constructs dense component matrices when a consumer requests them. A separate timing reports full materialization. The eight disagreements listed below all arise because the RDKit graph contains one or two additional short C$\cdots$C contacts. Independent reference bond tables are unavailable for these eight PCS2 molecules, so the comparison records policy concordance and leaves the contacts unscored. Accordingly, the fixed-graph result validates the cycle algorithms at matched topology, while the geometry-first result maps the domain in which the two disclosed perception policies agree.

Table S11: Geometry-first disagreements; ranks are ORACLE/RDKit.

Molecule	Extra RDKit C$\cdots$C (Å)	Rank	Ring sizes, ORACLE/RDKit
1-Ethynylbicyclo[1.1.1]pentane	1.8795	2/3	4,4 / 3,3,3
1-Ethynyltricyclo[2.2.0.0]hexane	1.8872; 1.8847	3/5	4,4,4 / 3×5
1-Methylbicyclo[1.1.1]pentane	1.8816	2/3	4,4 / 3,3,3
1-Methyltricyclo[3.2.0.0]hept-3-ene	1.8854	3/4	4,4,5 / 3,3,3,5
Spiro bicyclo[1.1.1]pentane–cyclopropane	1.8879	3/4	3,4,4 / 3×4
Spiro cyclopropane–tricyclo[2.2.0.0]hexane	1.8955; 1.8850	4/6	3,4,4,4 / 3×6
Tetracyclo[4.1.1.0.0]octane	1.8877; 1.8814	4/6	4,4,4,5 / $3 \times 5, 5$
Tetracyclo[4.3.0.0.0]nonane	1.8742	4/5	4,4,4,5 / 3,3,3,4,5

5.4 Stratified aromaticity comparison with RDKit

The aromaticity comparison uses 58 frozen repository geometries divided into nine strata: seven benzenoid, eight monocyclic heteroaromatic, nine fused, seven fused heteroaromatic, two nonbenzenoid, two boundary and 12 nonaromatic control systems, supplemented by six aromatic and five nonaromatic geometry-robustness cases. The boundary set includes cyclopropenone, whose three-membered ring has a substantial aromatic charge-separated contribution,[S13] and acenaphthylene, for which the five-membered ring is only weakly and paratropically aromatic.[S14] Every case passes a numerical gate requiring finite coordinates and a minimum all-atom distance above 0.55 Å.

Two comparison tracks separate connectivity perception from aromaticity assignment. In the geometry-first track, both programs receive the same elements and Cartesian coordinates and independently infer connectivity. In the fixed-graph track, RDKit receives the accepted nonmetal covalent graph before bond-order determination and sanitization. Exact equality of the one-based aromatic-atom sets is the primary endpoint; Jaccard overlap is also retained.

Table S12: Stratified aromaticity conformance against RDKit.

Track	Cases	Exact sets	Fraction (95% CI)	Mean/min. Jaccard
Geometry first	58	58	1.000 (0.938–1.000)	1.000/1.000
Fixed graph	58	58	1.000 (0.938–1.000)	1.000/1.000

Agreement is exact in every stratum and in both tracks. RDKit serves as an independent

comparator, and the result measures concordance over the declared corpus. The complete atom sets, category counts, geometry gates, source hashes, software versions and commit `0246d73` are frozen in `validation/oracle_rdkit_aromaticity_corpus.json`; the generating script is `validation/benchmark_oracle_rdkit_aromaticity_corpus.py`.

6 Overlap-constrained LCB25 assembly of [6]helicene

The construction starts from the PubChem [6]helicene connectivity

```
C1=CC=C2C(=C1)C=CC3=C2C4=C(C=C3)C=CC5=C4C6=CC=CC=C6C=C5
```

and uses a single library geometry,

```
LCB25(dataset=HPCS2, molecule=naphthalene)
```

which contains 10 carbon and 8 hydrogen atoms. Every placement therefore obeys the literal 20-atom ceiling, including hydrogens, while the complete target contains 26 heavy atoms and 16 hydrogens.

The deterministic beam search ranks partial covers by coverage, overlap, query size and source/match order, retaining 64 states. The seed may have no overlap, but every later candidate must add at least one target atom and overlap at least three already placed, non-collinear heavy atoms. Proper rotations are obtained by Kabsch alignment based on singular-value decomposition (SVD);[S15] reflections are rejected. A joint rigid-pose fit subsequently optimizes all selected placements to a 10^{-5} Å tolerance.

Table S13: [6]Helicene assembly record; atom indices are one based.

Step	Match	Overlap	New target atoms	Overlap RMSD (Å)
1	0	none	1–10	0
2	1	4,5,7,8,9,10	11,12,13,14	0.0138628
3	2	9,10,11,12,13,14	15,16,17,18	0.0127611
4	3	11,12,15,16,17,18	19,24,25,26	0.0127216
5	4	17,18,19,24,25,26	20,21,22,23	0.0105915

The joint pose fit converges in 18 iterations with maximum overlap RMSD 0.0138628 Å, below the strict 0.35 Å gate, and no chemical-quality warning is emitted. The initially assembled fused aromatic system is nearly planar and places the terminal rings in steric conflict. A deterministic displacement along the least-variance Cartesian normal breaks this saddle. SWITCH adds hydrogens and the analytic internal steric-release pass removes severe nonbonded contacts before the selected GFN-FF relaxation. The default 42-atom output is strict XYZ; `--embed-assembly-section` optionally appends the enriched section. The adjacent `oracle.xyz.assembly.v1` manifest records source–target mappings, shared and new atoms, per-atom provenance, beam-search diagnostics, pose convergence and the quality gate. Independent ORACLE preprocessing identifies 47 covalent bonds, six rings and C_2 symmetry. This is a starting geometry and construction validation, not a high-level equilibrium structure or conformer-ranking result.

The same procedure could be extended, without relaxing the 20-atom ceiling, by adding:

- pyridine, pyridazine, pyrimidine and pyrazine for azahelicenes;
- indole, benzofuran and benzothiophene overlap tiles;
- cyclohexane, cyclohexene and carbonyl-bearing ring tiles;
- compact cis- and trans-decalin junction motifs for steroid and hormone frameworks;
- five/six-membered heteroring junctions with explicit charge and multiplicity provenance.

Each candidate would still have to expose a reusable non-collinear overlap of at least three heavy atoms. A tile connected through only one bond would leave its relative orientation underdetermined and is therefore unsuitable for this construction.

7 LCB26 carbohydrate subset and glucose construction

Sugars were not represented in the published LCB25 library.[S6] LCB26 extends that chemical domain with higher-level optimized structures of representative sugars and their fragments. The carbohydrate-focused set comprises two D-xylose conformers; two erythrose, one erythrulose and three threose geometries; two fructose conformers; and the smaller oxygenated records glycidaldehyde, two glycidols, glycolic acid, glycolaldehyde, three glyceraldehydes, five glycerols and 1,3-dihydroxyacetone. The two source geometries carrying the same glyceraldehyde `c,g+` label are retained as separate occurrences because their Cartesian structures differ. Every record has a source hash and an ORACLE topology audit. The collection targets reusable ring, polyol, carbonyl and hydroxymethyl environments; it is not a claim of exhaustive carbohydrate coverage.

The α-D-glucopyranose test uses D-xylose `aconf10` as the ring-rich seed and glycerol `C5` as the hydroxymethyl-extension donor. The seed covers ten target heavy atoms. The second placement shares four non-collinear heavy atoms and contributes the two missing target atoms, giving complete 12/12 heavy-atom coverage. The joint pose optimization converges in one iteration with maximum overlap RMSD 0.334083 Å, below the 0.35 Å gate. GFN-FF relaxation preserves all specified stereocentres; the geometric, collision and stereochemical checks emit no warning. ORACLE reimport identifies 24 covalent bonds including hydrogens, one six-membered ring, and a valid C_1 persistent state.

Table S14: Overlap construction of α-D-glucopyranose.

Step	PPCS2 source	Covered	Shared	New
1	D-xylose `aconf10`	10	0	10
2	Glycerol `C5`	6	4	2

This example illustrates how ORACLE can use the emerging LCB26 chemical coverage to formulate a traceable starting hypothesis from validated local geometries. The constitu-

tional formula is supplied through an isomeric SMILES, after which the procedure selects, places, and audits the contributing fragments. The resulting strict XYZ can be opened directly in ORACLE or Avogadro, while the adjacent `oracle.xyz.assembly.v1` manifest records source–target mappings, shared and new atoms, per-atom provenance, beam-search diagnostics and the quality decision.

7.1 Conformer-specific leave-one-identity-out test

A blind validation removes 1,3-dihydroxyacetone `c,c` from the donor pool and rejects any remaining record with the same constitutional identity. The target `OCC(=O)CO` connectivity and the `c,c` conformer label are fixed in the protocol before reconstruction. The hidden reference coordinates are not read until the assembled structure and an independent no-fragment baseline have both been generated. ORACLE selects the maximum common heavy-atom subgraph of the acyclic erythrulose record; no 1,3-dihydroxyacetone geometry contributes to placement or relaxation.

The two H–O–C–C torsions define the conformational check. Both are 0° in the withheld reference structure. After local X–H-frame transfer from the selected fragment and analytic steric release they are +15.781° and +17.722°, so both retain the predeclared cis assignment. The comparison uses all graph-isomorphic atom maps, including terminal-group exchange, and reports the lowest proper-rotation Kabsch RMSD. The independent no-fragment comparison is the lowest-MMFF94-energy member of 20 deterministic ETKDGv3 starts and is used only as a disclosed external baseline; reference coordinates are not used to select it.

The fragment-seeded result reduces the all-atom RMSD by 8.2% and the heavy-atom RMSD by 13.4%, while its bond RMS error is marginally larger. The measurable benefit in this case is the closer hydroxyl orientation; the carbon skeleton and bond lengths retain similar accuracy. Both reconstructed structures show residual deviations from the exact C_{2v} symmetry of the reference structure: ORACLE assigns C_1 to the fragment-seeded geometry and C_2 to the baseline. These symmetry assignments make the residual distortion explicit

Table S15: Blind reconstruction of 1,3-dihydroxyacetone `c,c`.

Metric	LCB26 reconstruction	No-fragment baseline
Heavy-atom RMSD (Å)	0.06661	0.07689
All-atom RMSD (Å)	0.14405	0.15699
Bond RMS error (mÅ)	14.03	13.08
Maximum bond error (mÅ)	32.45	21.12
H–O–C–C torsions (deg)	$+15.78, +17.72$	$-25.32, -25.32$
Conformer assignment	`c,c`	`c,c`
ORACLE point group	C_1	C_2

and prevent the correct `c,c` rotamer label from being mistaken for an exact symmetry recovery. This withheld molecule provides a focused validation example, while database-wide error calibration rests on the broader corpus.

8 Hydrogen-bond and fragment audit

The archived water dimer contains two covalent components, atoms 1–3 and 4–6. Starting only from the element labels and Cartesian coordinates, ORACLE perceives the interfragment O1–H2$\cdots$O4 contact and keeps it distinct from the covalent edge set. Its numerical and serialized outputs are summarized below.

Table S16: Water-dimer interaction and fragment contract.

Quantity	ORACLE result
Covalent fragments	{1,2,3}; {4,5,6}
Donor–H$\cdots$acceptor	O1–H2$\cdots$O4
H$\cdots$O distance	1.840000 Å
D–H$\cdots$A angle	180.000°
Smooth contact weight	0.9999351064
Interaction PIC	primitive 5, H2–O4
Calibrated PL1 target	+0.02042577 Å

All four assertions pass: two connected covalent fragments, one interfragment hydrogen bond, one matching hydrogen-bond PIC and one calibrated PL1 selected-pair target. General hydrogen-bond perception accepts compatible N, O, P and S donors and acceptors; only the displayed O–H$\cdots$O(H) correction is claimed to be calibrated. Intramolecular contacts

close pseudocycles through the covalent path, while connected components, canonical ring domains, interaction domains and synthon clusters remain separately addressable fragment views.

8.1 Intramolecular hydrogen-bond resolution ladder

Nine frozen geometries extend the water-dimer contract from one intermolecular contact to peptide-like and oxygen-rich intramolecular environments. The C5 and C7 N-acetylglycinamide coordinates are the best-estimated structures reported for the glycine-dipeptide analogue,[S16] and the carbohydrate geometries are frozen LCB26 records; all are examined without further optimization.

Table S17: Hydrogen-bond resolution ladder across interaction definitions.

System	Atoms	Accepted	Closures	PIC rank	Contact-weight range
Glycine I	10	0	1	24/24	—
Glycine II	10	1	2	24/24	0.999920
N-acetylglycinamide C5	16	0	1	42/42	—
N-acetylglycinamide C7	16	1	1	42/42	0.999522
Dihydroxyacetone (c,c)	12	2	2	30/30	0.999376
α-D-xylose (00)	20	1	6	54/54	0.997311
α-D-xylose (10)	20	1	6	54/54	0.998159
D-fructose (34)	24	3	9	66/66	0.995883–0.998375
D-fructose (79)	24	4	7	66/66	0.075208–0.996603

The audit separates three complementary levels of information. Distances, angles, and the reported contact weight provide continuous evidence; declared thresholds turn that evidence into a deterministic contact list and interaction-closure graph; and primitive completeness is finally checked against those accepted decisions. All nine systems recover the full $3N-6$ rank. The low-weight contact in fructose conformer 79 remains explicit in the record, allowing a downstream application to adopt a stricter policy without repeating perception.

The examples resolve chemically different patterns. Glycine and N-acetylglycinamide each provide a paired comparison in which one conformer contains only a weak closure and the other contains an accepted O–H$\cdots$N or N–H$\cdots$O contact. In dihydroxyacetone, two hydroxyl donors address the same carbonyl oxygen. The two xylose structures reverse the

dominant O13/O15 donor–acceptor direction. Fructose supplies an extended multi-contact network. The nine-system set covers contact perception and primitive-rank preservation across these interaction patterns.

8.2 High coordination without organic-valence typing

Synthetic FeF_n spheres at a fixed 1.60 Å radius test the representation at accepted central coordination numbers 4, 6, 8, 10 and 12. The same code path gives continuous coordination numbers 3.986756, 5.980135, 7.973513, 9.966891 and 11.960269, respectively. The corresponding central synthons are serialized in `validation/perception_scope_audit.json`; no coordination-specific atom type or upper-valence branch is introduced.

For the 12-coordinate sphere, a radial scan from 1.45 to 2.25 Å leaves the directions unchanged and varies only the Fe–F radius. The continuous coordination number decreases smoothly and monotonically from approximately 12 to zero. A separate table-coverage assertion confirms that the covalent-radius layer required by this geometrical perception is available for atomic numbers 1–118. This check establishes periodic-table coverage of the representation and perception machinery. CM5/Mayer values from L0 or another declared source populate the electronic components of the synthon.

8.3 L0 feedback and open-shell spin observables

The feedback step uses the same PBE0/def2-TZVP L0 contract as the population examples. At the initial perceived geometry, population-analysis interface evaluates CM5 charges and the unrestricted total Mayer matrix

$$B_{AB} = 2 \sum_{\mu \in A, \nu \in B} \left[(P^{\alpha} S)_{\mu\nu} (P^{\alpha} S)_{\nu\mu} + (P^{\beta} S)_{\mu\nu} (P^{\beta} S)_{\nu\mu} \right].$$

The parallel spin-selective quantity replaces the sum by an $\alpha - \beta$ difference. A Hirshfeld stockholder partition of $\rho_{\alpha} - \rho_{\beta}$ supplies the complementary atomwise spin population. The

latter sums to $N_\alpha - N_\beta$; the spin Mayer matrix is symmetric, vanishes for a restricted closed shell and changes sign under exchange of the spin labels.

Eighteen neutral doublets span nine electronic classes: localized, unsaturated and conjugated cyclic carbon radicals, heteroatom-substituted carbon radicals, aryloxy, carbonyl, nitroxide, nitrogen-centered and peroxy systems. The archived record contains the initial geometry, CM5 vector, atomwise Hirshfeld spin populations, total and $\alpha - \beta$ Mayer values on every perceived covalent bond, SCF convergence, $\langle S^2 \rangle$ and immutable source hashes. These L0 observables are then attached to the persistent state before geometry regeneration or refinement. Re-perception of that geometry is the gate for the production calculation; the L0 state is not silently substituted for its final density. Results and the generating protocol are stored in `validation/l0_spin_population_cases_expanded.json` and `validation/run_l0_spin_population_cases.py` and `validation/run_additional_open_shell_cases.py`.

Table S18: L0 open-shell population audit across the declared electronic classes.

Doublet	$\langle S^2 \rangle$	Participation	max $\lvert s_A \rvert$	max $\lvert B^{\alpha-\beta} \rvert$
Allyl	0.7865	2.58	0.5509	0.0021
Benzyl	0.7856	3.26	0.6041	0.0358
Cyclopentadienyl	0.7715	3.22	0.4442	0.1655
Dimethyl nitroxide	0.7554	2.48	0.5077	0.4634
Ethyl	0.7550	1.70	0.7593	0.0306
Formyl	0.7545	1.98	0.6463	0.4494
Furanyloxy	0.7756	3.83	0.3190	0.2127
Hydroperoxyl	0.7542	1.70	0.7128	0.4081
Methyl	0.7547	1.38	0.8476	0.0063
Naphthyl	0.7695	1.89	0.8184	0.0442
Phenoxy	0.8046	4.85	0.3786	0.1860
Phenyl	0.7605	1.76	0.7847	0.0471
Propargyl	0.7740	2.26	0.5985	0.0566
Tert-butyl	0.7551	2.35	0.6411	0.0180
Vinyl	0.7641	1.43	0.8306	0.0358
Cyanomethyl	0.7702	2.05	0.6390	0.0390
Glycine radical	0.7561	2.96	0.5327	0.2660
Nitrogen dioxide	0.7694	2.69	0.4128	0.1280

The spin populations sum to 1.000000 in all 18 cases. The corpus mean absolute-spin

participation is 2.46 atoms with a bootstrap 95% interval of 2.09–2.89; the corresponding interval for the mean maximum absolute $\alpha - \beta$ Mayer component is 0.0781–0.2191. The nine-class stratification and corpus-level intervals quantify the range and reproducibility of the declared open-shell set; the two observations per class provide a descriptive within-class comparison rather than an independent statistical sample. Phenyl provides a particularly direct aromaticity–spin separation: its six ring Mayer orders lie between 1.408 and 1.514, whereas the absolute spin-selective ring components remain below 0.0471. Total Mayer equalization can therefore support the aromatic conjugation update while the $\alpha - \beta$ map retains the localized open-shell information.

8.4 State-specific electronic observables

The same ORACLE-perceived geometries were evaluated for distinct electronic states to test whether the persistent state can retain state-dependent electronic evidence. CH_2 was treated as both a singlet and a triplet, and O_2 as a triplet. For formaldehyde, a TDA/PBE0 singlet excitation was compared with the PBE0 ground state using the same Cartesian geometry and perceived covalent graph. The CH_2 C–H Mayer order changes from 0.938 in the singlet to 0.951 in the triplet, while the carbon CM5 charge changes from -0.179 to -0.201 e. The formaldehyde S_1 state lies 3.98 eV above the ground state in this TDA calculation; its C and O CM5 charges change from $+0.053$ and -0.290 e to -0.307 and $+0.003$ e, respectively, and the C=O Mayer order decreases from 2.098 to 1.445. These shifts show why state-specific CM5/Mayer observables can alter any subsequent atom typing, charge assignment or force-field parameter selection that consumes them. For uracil, the same comparison gives a TDA/PBE0 S_1 excitation at 3.64 eV. The two carbonyl Mayer orders change from 2.072 and 2.037 in S_0 to 1.966 and 1.828 in S_1, respectively; the latter change is particularly pronounced. An additional S_0 comparison with aqueous C-PCM changes the same two orders to 2.049 and 1.977 and shifts the CM5 charge vector. The C4=O4 carbonyl is more strongly perturbed in both comparisons, showing how the persistent electronic state can support solvent-aware

parameter selection when equilibrium bond parameters depend on the local bond order. The complete state comparison is stored in `validation/oracle_triplet_excited_cases.json`; the gas/aqueous comparison is stored in `validation/oracle_uracil_pcm.json`. The records were generated by the corresponding scripts on the ORACLE remote host.

8.5 Functional and basis-set sensitivity

To quantify the dependence of the electronic closure on the declared L0 level, the same ORACLE-perceived geometries were evaluated with PBE0/def2-SVP, PBE0/def2-TZVP, and B3LYP/def2-TZVP. The latter two calculations also used TDA for the first singlet excited state of formaldehyde and uracil. The complete charge vectors, bond lists, excitation energies, and source hashes are archived in `validation/oracle_method_basis_sensitivity.json`.

Table S19: Selected functional/basis sensitivity results at fixed ORACLE geometries. Mayer values are dimensionless and charges are in e.

Case	Observable	PBE0/SVP	PBE0/TZVP	B3LYP/TZVP
CH_2 S_0	C–H Mayer; q_C	0.944; −0.186	0.938; −0.179	0.936; −0.175
CH_2 T_1	C–H Mayer; q_C	0.969; −0.202	0.951; −0.201	0.951; −0.198
H_2CO S_0	C=O Mayer; q_C	2.106; +0.044	2.098; +0.053	2.092; +0.057
H_2CO S_1	C=O Mayer; q_C	1.459; −0.313	1.445; −0.307	1.430; −0.305
Uracil S_0	two C=O Mayer values	2.135/2.067	2.072/2.037	2.081/2.040
Uracil S_1	two C=O Mayer values	2.042/1.881	1.966/1.828	1.976/1.825

The formaldehyde TDA excitation energies are 3.99, 3.98, and 3.97 eV for PBE0/def2-SVP, PBE0/def2-TZVP, and B3LYP/def2-TZVP, respectively; the corresponding uracil values are 3.60, 3.64, and 3.52 eV. Across these cases, functional and basis changes perturb the observables, while the state change remains the dominant effect: the formaldehyde C=O Mayer order falls by about 0.65 from S_0 to S_1, and both uracil carbonyls decrease in S_1. This establishes the sensitivity range of the operational L0 contract without treating any single functional/basis pair as chemically exact.

8.6 State-blind control

The state-blind control holds the nuclear geometry and covalent graph fixed and assigns no state-dependent CM5, Mayer, or spin observables. Its structural output is consequently unchanged by construction. At PBE0/def2-TZVP, ORACLE gives a carbon CM5 change of -0.022 e and a mean C–H Mayer change of $+0.013$ for CH_2 singlet to triplet. For formaldehyde S_0 to S_1, the carbon and oxygen CM5 changes are -0.360 and $+0.293$ e, and the C=O Mayer change is -0.652; the two uracil carbonyl changes are -0.105 and -0.209. This control isolates the scientific consequence of retaining the electronic state at fixed geometry; it does not claim that persistence alone necessarily produces a divergent result. The machine-readable record is stored in `validation/oracle_state_blind_control.json`.

8.7 L0 CM5 hydrogen-bond response calibration

All calculations use fixed geometries. The design spans four donor types (O–H, N–H, P–H and S–H) and five acceptor types (O, N, carbonyl O, P and S). Each donor class contains four distinct neutral molecular environments and each acceptor class contains four; their complete within-class Cartesian products yield 320 directed dimers, or $N = 16$ observations for each of the 20 rules. The molecular series includes water, three alcohols, ammonia and alkylamines, four carbonyl environments, four phosphines and four sulfur hydrides or thiols. Nitrogen- and oxygen-acceptor contacts use $r_{H\cdots A} = 1.850$ Å; for sulfur and phosphorus acceptors this distance is scaled by the corresponding UFF van der Waals radius sum, giving 2.005 and 2.037 Å, respectively. In all cases $\angle$DHA $= 180.0°$, and each monomer and dimer received one PBE0/def2-TZVP single point followed by population-analysis interface CM5/Mayer analysis. The $H_2CO\cdots H_2CO$ C–H$\cdots$O structure is retained only as a weak control and is excluded.

The automated chemical audit confirms that every dimer contains the intended strong interfragment contact, no unintended strong contact, chemically acceptable intramolecular distances, and no interfragment clash. Within each chemical class, an unweighted arithmetic

mean combines the charge transfer with the separately neutral donor- and acceptor-local displaced charges. The parent canonical response supplies the atomwise shape, which is rescaled to the corresponding mean displaced charge while retaining an exactly zero sum. Table S20 collects the resulting 20 rules together with their signed canonical vectors, sample counts, and uncertainty estimates. For each scalar response, the archive reports the sample standard deviation, standard error, a deterministic 20 000-resample bootstrap 95% confidence interval and the largest leave-one-donor- or acceptor-environment-out shift. The complete 320-observation record remains machine readable in `validation/hbond_charge_response_averaged_l0.json`.

Using one rigid geometrical protocol isolates electronic substitution effects and makes the averaging reproducible. The dispersions and confidence intervals therefore describe variation among the represented neutral substitution environments at fixed angle, distance and conformation. The balanced 4×4 design within every rule also permits a direct leave-one-environment-out stability audit while the public 4×5 chemical table remains unchanged.

Across the 20 cells, the sample standard deviation spans 0.00184–0.01425 e for donor-local polarization, 0.00207–0.00844 e for acceptor-local polarization and 0.00466–0.01679 e for charge transfer. The largest donor spread occurs for N–H$\cdots$O=C and the largest charge-transfer spread for S–H$\cdots$N, identifying the classes most sensitive to substitution without destabilizing their means. The corresponding bootstrap 95% half-widths span 0.00087–0.00673, 0.00099–0.00396 and 0.00220–0.00794 e, respectively. Reporting both levels avoids two opposite errors: treating chemically meaningful within-class variation as numerical noise, or treating the mean of a balanced class as if it carried the full dispersion of an individual environment. Together with the maximum 0.0080 e bootstrap half-width and 0.0085 e leave-one-environment-out shift, these distributions establish that the 20 class means are stable at the resolution supported by this factorial design and available as executable CM5 hydrogen-bond charge-response parameters.

For the water monomer in O,H,H order, the L0 CM5 values are

$$\mathbf{q}_{\mathrm{mon}} = (-0.674181,\ +0.337164,\ +0.337016)\ e.$$

These are the raw electronic populations retained for provenance. The two hydrogens are symmetry equivalent, so the resident water response uses their mean and the exactly neutral vector $(-0.674180563, +0.337090282, +0.337090282)e$. The unrelated HF/STO-3G water fixture used to test population-analysis interface parsing is not a production parameter. For the averaged O–H$\cdots$O rule, the reference Mayer order is 0.06536 [0.06241, 0.06873], the donor- and acceptor-local displaced charges are 0.03199 [0.03011, 0.03368] and 0.03134 [0.02866, 0.03339] e, and the signed charge transfer is -0.06024 [-0.06389, -0.05583] e; brackets denote bootstrap 95% confidence intervals. The sign convention stores $\Delta Q_{\mathrm{CT}} < 0$ when the donor fragment becomes more negative.

All CM5 vectors in this validation are attached to their fixed reference geometries. Whenever ORACLE perceives a hydrogen bond in such a geometry, it first evaluates the tabulated unit vector and the reference strength s_c^{ref}, then stores $\mathbf{q}^0 = \mathbf{q}^{\mathrm{CM5}} - \sum_c s_c^{\mathrm{ref}} \Delta\mathbf{q}_c$. Re-evaluation at the same geometry reconstructs the original CM5 vector; changing the geometry adds only $\sum_c [s_c(\mathbf{R}) - s_c^{\mathrm{ref}}]\Delta\mathbf{q}_c$. The regression suite checks charge conservation, exact reference round trip, the zero-contact limit and the absence of a duplicated hydrogen-bond contribution.

Two frozen α-D-xylose conformers, excluded from the averages, provide an external intramolecular transfer test. Within each pair, changing only the donor O–H torsion removes the accepted O–H$\cdots$O contact while preserving all heavy atoms, the O–H length, and the C–O–H angle. The mean direction cosine between the observed and tabulated changes on D,H,A is 0.730. The class-mean rule gives a residual root-mean-square of 0.00896 e for an observed change of 0.00463 e: it captures part of the direction and provides an independent measure of the difference between intermolecular class averaging and a conformer-specific intramolecular response. The production route therefore uses the conformer-specific CM5

vector as its operational reference and the class table for the continuous change away from that reference. Inputs, charge differences and predictions are stored in `validation/sugar_oh_o_transfer_audit.json`.

The operational reference for each sugar is its own CM5 vector and frozen geometry. For every accepted O–H$\cdots$O contact, only the atoms in the declared donor, acceptor and charge-transfer supports are modified during the inverse step. Their summed $s_c(\mathbf{R}_{\mathrm{ref}})\Delta\mathbf{q}_c$ contributions are subtracted before the intrinsic vector is stored. Atoms outside all supports retain their reference CM5 values. Reapplying the same contacts reconstructs the complete CM5 vector exactly, whereas removal of a contact returns only its involved atoms toward the inferred no-hydrogen-bond values.

Table S20: Complete resident fixed-geometry hydrogen-bond charge-response parameters averaged over chemically equivalent dimers. $\Delta\mathbf{q}_D^{\mathrm{pol}}$ and $\Delta\mathbf{q}_A^{\mathrm{pol}}$ are separately neutral local vectors; $\Delta Q_{\mathrm{CT}} < 0$ denotes net transfer to the donor. Brackets report deterministic 20 000-resample bootstrap 95% confidence intervals for the class means. The symbol s gives the sample standard deviation across the 16 substitution environments in each class. Distances are in Å and all charge quantities in e.

Donor	Acceptor	N	r_{ref}	Mayer	$\Delta\mathbf{q}_D^{\mathrm{pol}}$	$\Delta\mathbf{q}_A^{\mathrm{pol}}$	ΔQ_{CT}
O–H	O	16	1.850	0.06536 [0.06241, 0.06873] $s = 0.00661$	O $+0.03199$; H$_1$ -0.03199 $P_D = 0.03199$ [0.03011, 0.03368]; $s = 0.00382$	O -0.03134; H$_1$ $+0.01595$; H$_2$ $+0.01539$ $P_A = 0.03134$ [0.02866, 0.03339]; $s = 0.00504$	-0.06024 [-0.06389, -0.05583] $s = 0.00867$
O–H	N	16	1.893	0.08312 [0.07825, 0.08827] $s = 0.01057$	O $+0.02866$; H$_1$ -0.02866 $P_D = 0.02866$ [0.02607, 0.03110]; $s = 0.00535$	N -0.04217; H$_1$ $+0.01405$; H$_2$ $+0.01421$; H$_3$ $+0.01391$ $P_A = 0.04217$ [0.03970, 0.04452]; $s = 0.00514$	-0.05619 [-0.06156, -0.05046] $s = 0.01172$
O–H	C=O	16	1.850	0.05565 [0.05427, 0.05721] $s = 0.00312$	O $+0.03594$; H$_1$ -0.03594 $P_D = 0.03594$ [0.03438, 0.03742]; $s = 0.00322$	C $+0.03462$; O -0.03462 $P_A = 0.03462$ [0.03280, 0.03670]; $s = 0.00418$	-0.07151 [-0.07415, -0.06890] $s = 0.00553$
O–H	P	16	2.037	0.10498 [0.09932, 0.11077] $s = 0.01217$	O $+0.05586$; H$_1$ -0.05586 $P_D = 0.05586$ [0.05337, 0.05825]; $s = 0.00520$	P -0.03785; H$_1$ $+0.01214$; H$_2$ $+0.01285$; H$_3$ $+0.01285$ $P_A = 0.03785$ [0.03636, 0.03928]; $s = 0.00309$	-0.07306 [-0.07819, -0.06788] $s = 0.01074$
O–H	S	16	2.005	0.07591 [0.07202, 0.08001] $s = 0.00841$	O $+0.05757$; H$_1$ -0.05757 $P_D = 0.05757$ [0.05537, 0.05982]; $s = 0.00464$	S -0.03032; H$_1$ $+0.01453$; H$_2$ $+0.01578$ $P_A = 0.03032$ [0.02848, 0.03187]; $s = 0.00356$	-0.07792 [-0.08233, -0.07351] $s = 0.00946$

Continued on next page

Table S20 (continued)

Donor	Acceptor	N	r_{ref}	Mayer	$\Delta\mathbf{q}_D^{\mathrm{pol}}$	$\Delta\mathbf{q}_A^{\mathrm{pol}}$	ΔQ_{CT}
N–H	O	16	1.850	0.06887 [0.06515, 0.07313] $s = 0.00848$	N $+0.02540$; H$_1$ -0.02540 $P_D = 0.02540$ [0.02083, 0.03054]; $s = 0.01022$	O -0.02341; H$_1$ $+0.01067$; H$_2$ $+0.01274$ $P_A = 0.02341$ [0.02085, 0.02613]; $s = 0.00563$	-0.05145 [-0.05481, -0.04826] $s = 0.00684$
N–H	N	16	1.893	0.08399 [0.07698, 0.09100] $s = 0.01475$	N $+0.01929$; H$_1$ -0.01929 $P_D = 0.01929$ [0.01365, 0.02536]; $s = 0.01236$	N -0.03522; H$_1$ $+0.00858$; H$_2$ $+0.01245$; H$_3$ $+0.01419$ $P_A = 0.03522$ [0.03220, 0.03849]; $s = 0.00670$	-0.04760 [-0.05298, -0.04166] $s = 0.01197$
N–H	C=O	16	1.850	0.05926 [0.05680, 0.06201] $s = 0.00550$	N $+0.03217$; H$_1$ -0.03217 $P_D = 0.03217$ [0.02597, 0.03943]; $s = 0.01425$	C $+0.02895$; O -0.02895 $P_A = 0.02895$ [0.02598, 0.03234]; $s = 0.00672$	-0.06489 [-0.06793, -0.06194] $s = 0.00630$
N–H	P	16	2.037	0.09802 [0.09047, 0.10594] $s = 0.01642$	N $+0.03916$; H$_1$ -0.03916 $P_D = 0.03916$ [0.03355, 0.04528]; $s = 0.01250$	P -0.02449; H$_1$ $+0.00202$; H$_2$ $+0.01124$; H$_3$ $+0.01124$ $P_A = 0.02449$ [0.02075, 0.02867]; $s = 0.00844$	-0.05162 [-0.05835, -0.04515] $s = 0.01387$
N–H	S	16	2.005	0.06885 [0.06312, 0.07498] $s = 0.01256$	N $+0.04348$; H$_1$ -0.04348 $P_D = 0.04348$ [0.03840, 0.04924]; $s = 0.01159$	S -0.01942; H$_1$ $+0.00453$; H$_2$ $+0.01489$ $P_A = 0.01942$ [0.01611, 0.02317]; $s = 0.00743$	-0.06045 [-0.06595, -0.05522] $s = 0.01134$
P–H	O	16	1.850	0.03619 [0.03319, 0.03931] $s = 0.00646$	P -0.01219; H$_1$ $+0.01219$ $P_D = 0.01219$ [0.01002, 0.01431]; $s = 0.00453$	O -0.00942; H$_1$ $+0.00807$; H$_2$ $+0.00134$ $P_A = 0.00942$ [0.00788, 0.01091]; $s = 0.00322$	-0.03002 [-0.03276, -0.02697] $s = 0.00609$

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Table S20 (continued)

Donor	Acceptor	N	r_{ref}	Mayer	$\Delta\mathbf{q}_D^{\mathrm{pol}}$	$\Delta\mathbf{q}_A^{\mathrm{pol}}$	ΔQ_{CT}
P–H	N	16	1.893	0.05102 [0.04620, 0.05587] $s = 0.01028$	P -0.02244; H$_1$ $+0.02244$ $P_D = 0.02244$ $[0.01846, 0.02695]$; $s = 0.00905$	N -0.01689; H$_1$ $+0.00424$; H$_2$ $+0.00632$; H$_3$ $+0.00632$ $P_A = 0.01689$ $[0.01585, 0.01793]$; $s = 0.00220$	-0.02161 [-0.02707, -0.01535] $s = 0.01236$
P–H	C=O	16	1.850	0.03078 [0.02889, 0.03271] $s = 0.00407$	P -0.00651; H$_1$ $+0.00651$ $P_D = 0.00651$ $[0.00464, 0.00850]$; $s = 0.00409$	C $+0.02130$; O -0.02130 $P_A = 0.02130$ $[0.01994, 0.02286]$; $s = 0.00310$	-0.05216 [-0.05442, -0.04995] $s = 0.00475$
P–H	P	16	2.037	0.09790 [0.09407, 0.10138] $s = 0.00774$	P -0.00569; H$_1$ $+0.00569$ $P_D = 0.00569$ $[0.00393, 0.00756]$; $s = 0.00380$	P -0.00227; H$_1$ -0.00684; H$_2$ $+0.00456$; H$_3$ $+0.00456$ $P_A = 0.00911$ $[0.00798, 0.01019]$; $s = 0.00232$	-0.01140 [-0.01487, -0.00796] $s = 0.00728$
P–H	S	16	2.005	0.04841 [0.04511, 0.05154] $s = 0.00679$	P $+0.00322$; H$_1$ -0.00322 $P_D = 0.00322$ $[0.00199, 0.00464]$; $s = 0.00282$	S $+0.00286$; H$_1$ -0.00592; H$_2$ $+0.00306$ $P_A = 0.00592$ $[0.00494, 0.00691]$; $s = 0.00207$	-0.02336 [-0.02636, -0.02023] $s = 0.00650$
S–H	O	16	1.850	0.04426 [0.04197, 0.04658] $s = 0.00484$	S -0.00256; H$_1$ $+0.00256$ $P_D = 0.00256$ $[0.00174, 0.00347]$; $s = 0.00184$	O -0.01852; H$_1$ $+0.01215$; H$_2$ $+0.00637$ $P_A = 0.01852$ $[0.01692, 0.02009]$; $s = 0.00339$	-0.04232 [-0.04438, -0.03998] $s = 0.00466$
S–H	N	16	1.893	0.06603 [0.06145, 0.07061] $s = 0.00959$	S -0.00682; H$_1$ $+0.00682$ $P_D = 0.00682$ $[0.00355, 0.01070]$; $s = 0.00759$	N -0.02387; H$_1$ $+0.01001$; H$_2$ $+0.00686$; H$_3$ $+0.00700$ $P_A = 0.02387$ $[0.02195, 0.02568]$; $s = 0.00395$	-0.02909 [-0.03640, -0.02051] $s = 0.01679$

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Table S20 (continued)

Donor	Acceptor	N	r_{ref}	Mayer	$\Delta\mathbf{q}_D^{\mathrm{pol}}$	$\Delta\mathbf{q}_A^{\mathrm{pol}}$	ΔQ_{CT}
S–H	C=O	16	1.850	0.03772 [0.03597, 0.03954] $s = 0.00374$	S $+0.00435$; H_1 -0.00435 $P_D = 0.00435$ [0.00308, 0.00552]; $s = 0.00258$	C $+0.02913$; O -0.02913 $P_A = 0.02913$ [0.02748, 0.03107]; $s = 0.00381$	-0.06024 [-0.06271, -0.05779] $s = 0.00513$
S–H	P	16	2.037	0.10920 [0.10338, 0.11453] $s = 0.01179$	S $+0.01824$; H_1 -0.01824 $P_D = 0.01824$ [0.01628, 0.02019]; $s = 0.00415$	P -0.02025; H_1 $+0.00472$; H_2 $+0.00777$; H_3 $+0.00777$ $P_A = 0.02025$ [0.01883, 0.02167]; $s = 0.00297$	-0.03949 [-0.04376, -0.03533] $s = 0.00898$
S–H	S	16	2.005	0.06307 [0.05970, 0.06671] $s = 0.00744$	S $+0.02313$; H_1 -0.02313 $P_D = 0.02313$ [0.02160, 0.02465]; $s = 0.00327$	S -0.01508; H_1 $+0.00607$; H_2 $+0.00900$ $P_A = 0.01508$ [0.01333, 0.01676]; $s = 0.00364$	-0.04793 [-0.05169, -0.04427] $s = 0.00789$

8.8 Water endpoint and pentamer regression

The water–water production response is not obtained by refitting the 4×5 dimer table. Water dimer observations remain useful inside the general O–H/O class average and its audit, but they never override the dedicated water route. Its production parameters are taken from Equations 18–20 of Ref. S3: the charge-transfer scale is $\beta = 0.10e$, while the published missing-coordination correction is adapted by imposing the isolated CM5 and saturated TIP3P-FB limits exactly. ORACLE's continuous hydrogen-bond strength replaces the original integer count.

The regression geometry is the symmetric pentamer with fixed water monomers and four H$\cdots$O contacts of unit strength. The central water donates through both hydrogens and accepts at both lone-pair capacities. Its two $+\beta$ and two $-\beta$ fragment-charge contributions cancel exactly; the remaining neutral polarization increment converts the CM5 reference into $(-0.84844, +0.42422, +0.42422)e$. The runtime regression reproduces these three charges within $2 \times 10^{-14}e$ and the central and total pentamer charges within the same tolerance. Thus the pentamer verifies the endpoint constraint and contributes no fitted parameter.

At a nonperiodic boundary, smooth exposure supplies the missing neutral polarization capacity toward the same endpoint, but no charge transfer, because no physical partner exists. This is the direct continuous generalization of the published boundary construction and avoids a self-consistent induced-charge iteration.

The covalent graph and donor/acceptor identities are compiled once. The single versioned 4×5 response table is installed in the common chemistry runtime, where it is available to ORACLE and to every optimization, exploration or simulation tool. The shared `prepare_cm5_hydrogen_bond_charge_model` entry point accepts connectivity, geometry and intrinsic or environment-polarized CM5 charges. It returns tuned charges together with separate polarization, charge-transfer, contact-class and unresolved-contact records. Trajectory and optimization steps reuse its skinned H$\cdots$A pair list without re-entering the complete perception workflow; the one-shot `tune_cm5_hydrogen_bond_charges` entry point applies the

identical contract to an individual geometry.

The overlap audit uses two independently evaluated water dimers sharing the central water. Atom-observation counts are $(1, 1, 1, 2, 2, 2, 1, 1, 1)$; the assembled trimer charges sum to 1.1×10^{-16} e and the largest minimum-displacement correction needed to enforce neutrality is 2.2×10^{-6} e. This is a deliberately small exact audit of the same fragment-assembly equations used for large targets.

9 Frozen Gly–Ala–Gly population synthesis

SWITCH converted the stereochemical SMILES string `NCC(=O)N[C@@H](C)C(=O)NCC(=O)O` once into a deterministic Cartesian seed, which was retained without optimization in a completely extended configuration. ORACLE perceives no intramolecular hydrogen bond in this reference, so all s_c^{ref} vanish and the CM5 vector assembled from the capped calculations is already $\mathbf{q}^0$. The response library is applied only if a later geometry forms a contact, at which point it adds $s_c(\mathbf{R})\Delta\mathbf{q}_c$ without changing the stored fragment ownership or repeating the electronic calculations. Every atom, including each completed hydrogen, is owned by one of three residue cores, Gly^1, Ala^2 or Gly^3. Each core is dilated by two covalent-graph shells. Bonds crossing only the resulting outer halo boundary are capped by hydrogen along their original direction, while all retained Cartesian coordinates remain unchanged. Each core-plus-halo fragment received one L0 PBE0/def2-TZVP single point and population-analysis interface CM5/Mayer analysis.

The three cores partition all 27 atoms exactly once, while their halos jointly cover all 26 covalent bonds. Only core charges enter the molecular assembly; repeated halo populations and cap mappings diagnose boundary sensitivity, and Mayer observations are taken from the most interior available fragment. After this selection, a minimum-displacement correction restores the molecular charge. Neutral local groups are then constructed for fast multipoles, without imposing neutrality on the overlapping QM fragments. Complete core/halo maps,

observation counts, boundary standard errors and the independent full-molecule comparison are recorded in `validation/gly_ala_gly_fragment_charges.json` and in the regenerated parallel Oracle record `validation/oracle_gly_ala_gly_fragment_qm_parallel.json`. The comparison isolates population error from finite electronic environments and assembly at fixed geometry and conformation. Its effect on downstream electrostatic energies, multipoles and fitted force-field parameters must be propagated against the tolerance of the intended application.

With this separation of ownership and context, the regenerated parallel Oracle synthesis requires a maximum neutrality correction of 3.8971×10^{-4} e. Against the direct frozen-molecule L0 result, its CM5 root-mean-square and maximum absolute errors are 0.0032 and 0.0101 e, respectively. The ownership rule therefore gives a quantitative accuracy benchmark for overlap assembly while the repeated halo observations retain a direct boundary diagnostic. Because the SWITCH Cartesian seed, every retained coordinate, and the PBE0/def2-TZVP/population-analysis interface protocol are unchanged, this improvement follows from the ownership rule under an unchanged geometry and electronic level. Bond lengths in the frozen 27-atom seed span 0.890–1.540 Å, and all 26 declared covalent bonds are covered by at least one fragment.

10 Fixed-geometry L0 metal-complex examples

Ferrocene and cisplatin test the L0 population contract in chemically distinct 3d and 5d regimes through fixed-geometry PBE0/def2-TZVP single points with tight SCF convergence followed by population-analysis interface CM5 and Mayer analysis. Both geometries remain fixed. The eclipsed D_{5h} ferrocene geometry is the resident ORACLE regression structure; Fe is treated all-electron. Cisplatin uses a fixed idealized C_{2v} cis-$[PtCl_2(NH_3)_2]$ geometry and the native scalar-relativistic def2 ECP on Pt. Thus the examples validate population coverage and symmetry equivalence within the stated fixed structures.

Table S21: Fixed-geometry L0 CM5 charges and metal–ligand Mayer orders.

System	Quantity	Mean	Range
Ferrocene	Fe charge (e)	+0.984411	—
	C charge (e), 10 atoms	−0.223184	$[-0.223185, -0.223183]$
	H charge (e), 10 atoms	+0.124743	$[+0.124743, +0.124743]$
	Fe–C Mayer order, 10 bonds	0.440973	$[0.440971, 0.440975]$
Cisplatin	Pt charge (e)	+0.657196	—
	Cl charge (e), 2 atoms	−0.453199	equivalent
	N charge (e), 2 atoms	−0.919969	equivalent
	H charge (e), 6 atoms	+0.348190	$[+0.347131, +0.350308]$
	Pt–Cl Mayer order, 2 bonds	0.802090	equivalent
	Pt–N Mayer order, 2 bonds	0.520231	equivalent

The converged electronic energies are −1650.274385802 and −1152.622373103 E_h for ferrocene and cisplatin, respectively; they are reported only as calculation fingerprints. The summed CM5 charges are 1.94×10^{-16} and -2.22×10^{-16} e. Full coordinates, atomic charges, Mayer orders, electronic requests and convergence flags are frozen in `validation/oracle_l0_metal_complexes.json`.

11 Finite-group and orientation audit

For every analytic geometry, the operation matrices and atom permutations were tested jointly for identity, closure, inverses, orthogonality and consistent composition. The point-group label and complete operation count were then recomputed after a rotation of 0.37 rad about z, a rotation of 0.83 rad about y, and translation by $(2.0, -3.0, 0.7)$ Å.

Table S22: Finite-group and rigid-orientation results.

Case	Group	Operations	$\epsilon_{\max}$ (Å)	Rotated result
Water	C_{2v}	4	8.66×10^{-17}	$C_{2v}/4$
Benzene	D_{6h}	24	2.19×10^{-15}	$D_{6h}/24$
Ferrocene, eclipsed	D_{5h}	20	4.88×10^{-15}	$D_{5h}/20$
Ferrocene, staggered	D_{5d}	20	1.51×10^{-15}	$D_{5d}/20$
Methane	T_d	24	0	$T_d/24$
Cubane	O_h	48	0	$O_h/48$
Icosahedral model	I_h	120	5.79×10^{-16}	$I_h/120$

The eclipsed-ferrocene case specifically guards against a complete but incorrect C_{5h} subgroup that a partial inertia-frame treatment could select after rotating a partially degenerate inertia frame.

For the nontrivial projection shown in the main text, two vertices of analytic cubane were displaced independently by deterministic vectors of magnitude $O(10^{-4})$ Å. At $\tau = 10^{-3}$ Å, ORACLE assigned O_h, recovered all 48 operation–permutation pairs, and projected the geometry with maximum and RMS atomic displacements of 3.2945×10^{-4} and 1.3758×10^{-4} Å, respectively. Strict 10^{-8} Å reanalysis again returned $O_h/48$ with maximum residual 9.74×10^{-16} Å. Input and projected coordinates are stored in `validation/nontrivial_symmetrization_audit.json`.

12 Quasi-symmetry threshold and Cartesian projection

Table S23: Absolute threshold sweep for distorted formaldehyde.

τ (Å)	Group	Operations	$\epsilon_{\max}/\tau$	$\delta_{\max}$ (Å)
1.0×10^{-4}	C_s	2	0.0021	0
2.0×10^{-4}	C_{2v}	4	0.8071	8.07×10^{-5}
5.0×10^{-4}	C_{2v}	4	0.3228	8.07×10^{-5}
1.0×10^{-3}	C_{2v}	4	0.1614	8.07×10^{-5}

The selection rule chooses the smallest stable plateau value for which $\epsilon_{\max}/\tau \leq 0.8$. It therefore selects 5.0×10^{-4} Åfor this structure. Reanalysis of the projected geometry at 10^{-8} Åreturns C_{2v} with maximum residual 3.09×10^{-15} Å.

The distributed quasi-symmetric water case provides a compact assignment and adjustment regression. Its two O–H vectors differ slightly and one H atom has a 2.0×10^{-4} Å out-of-plane component. ORACLE first assigns the complete operation–permutation set at the declared threshold, then applies the Cartesian group projector only when that set has the expected order. The serialized symmetry section distinguishes `APPLIED`, `NOT_REQUIRED_`

NUMERICAL_NOISE, SKIPPED_INCOMPLETE_GROUP and NOT_APPLICABLE_C1. This prevents an incomplete candidate subgroup from being used as a geometry projector.

13 Continuous-descriptor contracts

Six per-atom quantities were audited: continuous coordination number, Z^{eff}, charge, covalency, delocalization and strain. Rigid-motion and atom-order tests compare the complete per-atom matrix after restoring the original indexing. The local-response test moves one atom through nine equally spaced points over a 2.0×10^{-4} Åinterval and requires unchanged discrete topology throughout.

Table S24: Descriptor invariance and sampled local response.

Case	Rigid-motion maximum	Reordering maximum	Maximum adjacent change
Water	0	0	3.56×10^{-5}
Benzene	2.22×10^{-15}	7.63×10^{-16}	6.15×10^{-5}
Methane	9.99×10^{-16}	0	6.68×10^{-5}

These sampled changes establish the numerical regression contract in the neighborhood of the three reference geometries. Crossing a perception boundary remains a separate event outside this local test.

14 L1-to-PL1 and synthon atom-class checks

The posterior CV implementation is tested against the fitted second- and third-period line parameters and accepts the supported element set without extrapolation. This radius-aware period-line Gaussian is the only posterior calibration exposed by the implementation; no compressed or legacy variant can be selected.

The PL1 fragment regression uses two water monomers with oxygen atoms separated by 2.80 Å. ORACLE retains four covalent bonds, identifies two fragments and adds the 1.84 Å H$\cdots$O pseudo-bond. The contact lies in the calibrated O–H$\cdots$O(H) class and produces one

positive target larger than 0.02 Å. The same perception code labels whether every contact is intra- or intermolecular; uncalibrated N/O/P/S classes remain pseudo-bonds without PL1 parameters.

The bounded atom-class example uses the ethanol geometry distributed with the ORACLE validation archive. Complete-link clustering is performed separately for each element with maximum within-class ranges of 0.08 for Z^{eff} and charge, 0.04 for covalency and delocalization, 0.05 for strain, and 0.10 for each of the first- and second-π indices. The default result is

Threshold scale	C classes	O classes	H classes	Total
0.25	(1),(2)	(3)	(4–8),(9)	5
0.50	(1),(2)	(3)	(4–8),(9)	5
1.00	(1),(2)	(3)	(4–8),(9)	5
2.00	(1),(2)	(3)	(4–8),(9)	5
4.00	(1),(2)	(3)	(4–9)	4
8.00	(1–2)	(3)	(4–9)	3

Every reported class satisfies all seven declared range bounds. Reversing the input atom order and restoring the one-based identifiers reproduces the same partition. The complete records, centroids and maximum spreads are stored in `validation/synthon_atom_class_audit.json`. This is deliberately a perception and compression example, not a force-field parameterization.

15 Single-model electronic-observable comparison

ORACLE emits one atomic charge and one bond order per object. The paired quantum-mechanical (QM) level is selected only when both the Charge Model 5 (CM5) atomic charges and Mayer bond orders are complete;[S1,S2] otherwise the whole structure uses the

paired electronegativity/Pauling level. Partial vectors are never mixed. The saccharin diagnostic contains 17 atomic charges and all 18 perceived covalent bonds. The resulting Pearson/Spearman correlations are 0.862/0.891 for charge and 0.907/0.924 for bond order. Raw values, one-based bond pairs and the source-log SHA-256 hash are stored in `validation/oracle_observable_comparison_saccharin.json`; the figure is regenerated by

```
python validation/plot_observable_comparison.py
```

The comparison measures ordering and trend preservation without fitting any parameter. Whenever CM5 atomic charges and Mayer bond orders are available, they remain authoritative over the geometry-only estimates.

16 Saccharin correction and Hessian audit

The L1 equilibrium geometry was read from the archived converged checkpoint. The independent higher-level geometry is the final reference structure. ORACLE applied 18 transversal core–valence (CV) targets and 13 additional targets: six aromatic C–C corrections, two S–O and one S–N polar-sulfur residuals, and four legacy high-level C–H residuals. The S–O and S–N discrepancies are 3.148 and 3.471 mÅ, defining a common arithmetic midpoint of -3.310 mÅ. Its subtraction leaves balanced residuals of -0.162 and $+0.161$ mÅ. The C–H amplitude, -0.870 mÅ, is inherited from the original high-level C–H enhancement that is deliberately absent from the method-independent CV posterior. A direct saccharin scan places the bond-RMS minimum near -0.70 mÅ, consistent with the transferred value.

The complete combined operation reduces the all-bond RMS from 3.369 to 1.115 mÅ. The lower-level VPT2 increments used in this archived observable correction are

$$\Delta(A, B, C) = (-8.520, -4.656, -3.631)\ \text{MHz},$$

whereas the experimental ground-state constants are

$$(A_0, B_0, C_0) = (1298.4982, 871.2573, 579.46230)\ \text{MHz}.$$

The complete coordinates, bond-by-bond residuals, equilibrium and corrected constants, input SHA-256 fingerprints and plotting data are stored in

`validation/saccharin_structure_case.json`.

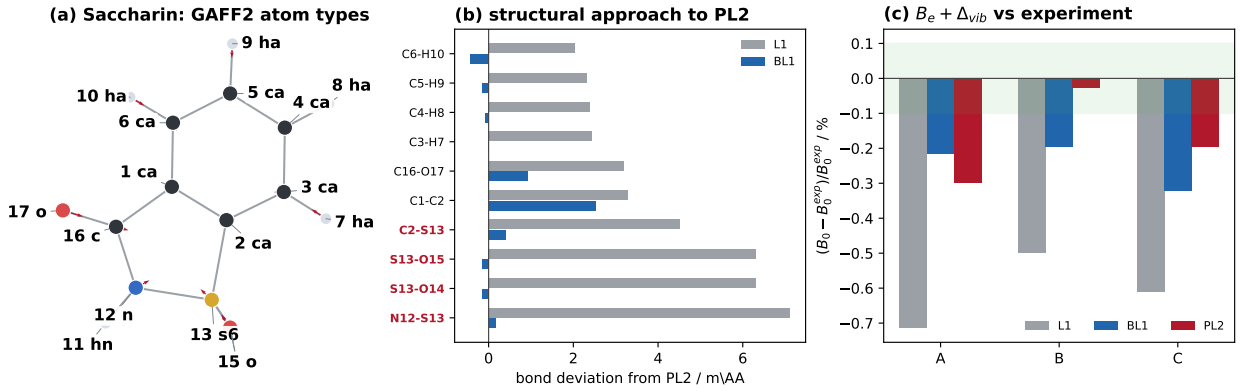


Figure S1: Saccharin structural-correction audit and final bond residuals.

The atom-typing audit uses the QM Hessian only after projection into ORACLE PICs. At the default thresholds it produces 13 atomic types, 101 primitive classes and 258 PIC–PIC couplings. Threshold scales 0.5, 1, 2 and 4 give 15, 13, 12 and 11 atomic types, respectively. The PIC-only Cartesian Hessian matrix RMS is 18.3367%, changing by less than 5.0×10^{-8} in absolute relative RMS across the sweep. This invariance establishes the stability of the atom-typing and PIC-coupling assignments across the tested threshold range; the full structural correction is quantified independently by the bond-RMS reduction reported above.

17 Saccharin-to-LCB25 parameter-donor retrieval

The frozen PCS2 ZIP archive contains 331 molecular XYZ members after excluding archive metadata. All were perceived successfully at the same geometry-only availability level as

the saccharin query. This common level gives ten query atom classes, nine bond classes and two cycles; it is intentionally distinct from the thirteen QM-level classes used in the Hessian audit. Gaussian similarities use the seven ORACLE synthon components normalized by the default class thresholds. Bond kernels additionally use endpoint orientation and the σ, π and π–π components; cycle kernels use size, element multiset, aromatic fraction and fused degree. Atom, bond and cycle coverages carry weights 0.45, 0.40 and 0.15.

The leading complete-molecule match, with combined similarity 0.628682, is 1H-quinolin-4-one, followed by 2-pyridinone (0.603114). 2-Thiouracil is the leading sulfur-containing entry among the first five (0.592398). A complementary six-molecule panel increases weighted parameter coverage from 0.628682 to 0.808259; sulfur dioxide is selected second because it supplies the best local S–O match despite ranking poorly as a whole molecule. The best S–O bond similarity is 0.534, whereas the best S–N and C–S similarities are only about 0.001 or smaller. These records quantify complementary parameter donors and provide the statistical selection criterion for library expansion.

The distinct joint sulfur-centred environment is the quantitative admission criterion for saccharin as the first LCB26 member. LCB26 contains the corresponding higher-level geometries, fingerprints and coverage evidence. The complete 331-entry ranking, per-class donor lists, metric definition, source hashes and greedy panel are stored in

`validation/saccharin_lcb25_retrieval.json`.

The positive control uses the 331 molecule names common to HPCS2 and PCS2. For each target, its PCS2 entry is removed, its HPCS2 geometry supplies the geometry-only query descriptors, and each covalent bond length is copied from the most similar bond belonging to a different PCS2 molecule. Retrieved bond lengths are imposed simultaneously by ORACLE PIC back-transformation. The reported selector is fixed to cyclic 6–30-atom S/N systems with mean donor similarity at least 0.80 and improvement over HPCS2; seven cases qualify, and thiazole has the lowest retrieved bond RMS. Its mean/minimum donor similarities are 0.908897/0.769542. The bond RMS against held-out PCS2 decreases from 14.148 to

3.191 mÅ, while aligned Cartesian RMS decreases from 0.014461 to 0.005149 Å. The back-transformation converges in two iterations with maximum residual 6.1×10^{-11}. Per-bond donor identities, similarities, transferred lengths, reconstructed coordinates, archive hashes and the 20 best leave-one-out cases are stored in `validation/lcb25_leave_one_out.json`.

18 SMILES–xTB–LCB25 differential geometry validation

The target is fixed as thiazole before calculation. Its initial geometry is generated from SMILES by SWITCH followed by GFN-FF. GFN2-xTB with tight optimization then supplies the low-level target geometry. The thiazole PCS2 record is excluded from retrieval and used only to evaluate the final result.

ORACLE constructs the complete redundant primitive state and matches each bond, angle, dihedral, and out-of-plane coordinate to another LCB25 PCS2 molecule using primitive identity and continuous effective-environment distance. Only the ten selected donor conformers are optimized with GFN2-xTB. For every supported coordinate, the PCS2–xTB difference of the donor is added to the target xTB value. Coverage is 8/8 bonds, 11/11 angles, 12/12 dihedrals, and 3/3 out-of-plane coordinates; the missing-fragment request list is empty.

Simultaneous reconstruction reduces bond RMS from 13.281 to 4.877 mÅ, angle RMS from 0.864 to 0.718°, and aligned Cartesian RMS from 0.016095 to 0.008617 Å. The frozen aromatic assignment identifies one six-electron ring and the frozen point-group state is C_s. Local-ring planarity and finite-group projection retain dihedral and out-of-plane RMS below 4×10^{-7} degrees. The full transfer provenance, xTB outputs, atom mapping, quality gates, primitive coverage, and error decomposition are stored in `validation/lcb25_xtb_differential.json`.

19 Serialized-state and software tests

Round-trip regressions reload the enriched geometry with perception disabled and verify unchanged topology, cycles, synthons, primitive definitions, symmetry matrices, atom permutations, thresholds, and Cartesian fingerprints. The human-readable report is regenerated from the same frozen state without mutating it. The ORACLE-specific audits summarized in the main text comprise 11 topology cases, three descriptor-invariance probes, seven finite-group algebra cases, seven rigid-orientation cases, four quasi-symmetry thresholds, one hydrogen-bond/fragment case, nine intramolecular interaction cases, five high-coordination cases, one 12-coordinate radial scan and one synthon-class audit; all supplied assertions pass. The overlap extension adds two direct tests: complete 42-atom [6]helicene construction with five bounded fragment placements, and literal rejection of the 18-atom naphthalene source when the maximum is set to 17. The carbohydrate extension adds a complete two-placement α-D-glucopyranose construction, a warning-free stereochemical quality gate and independent ORACLE reimport. Its leave-one-identity-out test fixes the target conformer in advance, verifies both defining torsions and compares the reconstruction with a deterministic no-fragment baseline.

20 Baker transition-state benchmark

The six representative Baker structures shown in the main text are drawn from the complete 25-case single-geometry benchmark. ORACLE perceives the transition-state evidence and coordinate construction performs the optimization. The table records the representative cases retained in the main text; the full corpus, source geometries, charge/multiplicity metadata, seed provenance and checksums are included in the accompanying data archive.

Table S25: Complete 25-case transition-state manifest.

ID	Structure	Rank	Coordinate-generation iterations	Optimization
1	HCN	3	10	accepted
2	HCCH	6	17	accepted
3	H_2CO	6	13	accepted
4	CH_3O	9	12	accepted
5	cyclopropyl	18	9	accepted
6	bicyclobutane	24	9	accepted
7	bicyclobutane	24	29	accepted
8	formyloxyethyl	24	7	accepted
9	parent Diels–Alder	42	18	accepted
10	s-tetrazine	18	6	accepted
11	trans-butadiene	24	4	accepted
12	ethane H_2 abstraction	18	11	accepted
13	HF abstraction	18	18	accepted
14	vinyl alcohol	15	20	accepted
15	HOCl	6	9	accepted
16	$H_2PO_4^-$	15	13	accepted
17	Claisen	36	11	accepted
18	silyene insertion	27	11	accepted
19	HNCCS	9	4	accepted
20	$HCONH_3^+$	15	9	accepted
21	acrolein rotation	18	4	accepted
22	HCONHOH	15	8	accepted
23	HCN + H_2	9	6	accepted
24	H_2CNH	9	10	accepted

ID	Structure	Rank	Coordinate-generation iterations	Optimization
25	$HCNH_2$	9	10	accepted

The complete machine-readable source for this table is `validation/baker_ts_full_manifest.tsv`; it also records the geometry, chart-definition, and geometry hashes for every case.

Table S26: Representative transition-state examples.

Case	TS topology	Rank	Coordinate-generation steps	Optimization
HCN (#1)	HCN→HNC isomerization/insertion	3	10/10	optimized
Cyclopropyl (#5)	ring opening	18	9/11	optimized
Bicyclobutane (#7, representative of two structures)	strongly distorted ring opening	24	9/8	optimized
s-Tetrazine (#10)	dissociation; near-TS seed	18	6/6	optimized
trans-Butadiene (#11)	torsional isomerization; 130° seed	24	4/5	optimized
Claisen (#17)	rearrangement	36	11/8	optimized

Across all 25 cases, the coordinate-generation and native Gaussian routes required 292 and 358 iterations, respectively, with medians of 10 and 11. Cases 10 and 11 use near-TS source geometries; these protocols condition the starting structures and leave the perception and coordinate-construction rules unchanged. For bicyclobutane, the table reports the representative #7 geometry; the second rank-24 bicyclobutane geometry is included in the 25-case aggregate but is not assigned a separate iteration pair in this representative table.

21 Reproduction commands

From the ORACLE manuscript source directory, with an arbitrary local ORACLE source checkout chosen by the user:

```
export ORACLE_ROOT=/path/to/oracle-source
export LCB25_ROOT="$ORACLE_ROOT/data/lcb25"
# Populate LCB25_ROOT from the published PCS2, HPCS2 and SE archives.
```

```
python validation/validate_topology_corpus.py "$ORACLE_ROOT"
python validation/validate_large_ring_systems.py "$ORACLE_ROOT"
python validation/validate_hbond_fragments.py "$ORACLE_ROOT"
python validation/validate_perception_scope.py "$ORACLE_ROOT"
python validation/build_perception_scope_figure.py
python validation/validate_descriptor_invariance.py "$ORACLE_ROOT"
python validation/validate_symmetry_groups.py "$ORACLE_ROOT"
python validation/probe_orientation_covariance.py "$ORACLE_ROOT"
python validation/validate_quasisymmetry.py "$ORACLE_ROOT"
python validation/validate_synthon_atom_classes.py "$ORACLE_ROOT"
python validation/plot_observable_comparison.py
python validation/build_validation_figure.py
python validation/build_saccharin_structure_case.py \
  "$ORACLE_ROOT" /path/to/saccharin_L1.xyzin /path/to/saccharin_reference.xyzin
python validation/build_saccharin_pic_case.py \
  "$ORACLE_ROOT" /path/to/saccharin_L1.xyzin \
  /path/to/saccharin_L1.log /path/to/saccharin_L1.fchk
python validation/build_saccharin_lcb25_retrieval.py \
  "$ORACLE_ROOT" /path/to/saccharin_L1.xyzin --lcb25-root "$LCB25_ROOT"
python validation/build_lcb25_leave_one_out.py \
  "$ORACLE_ROOT" --lcb25-root "$LCB25_ROOT"
python validation/validate_lcb25_xtb_differential.py \
  "$ORACLE_ROOT" --lcb25-root "$LCB25_ROOT" --xtb-executable xtb
python validation/benchmark_oracle_rdkit_rings.py \
  "$ORACLE_ROOT" --repeats 15
python validation/summarize_benchmark_uncertainty.py
python scripts/make_external_benchmark_figure.py
```

```
python tools/validate_lcb26_leave_one_out.py \
  --output-dir /path/to/output/dihydroxyacetone-leave-one-out
```

The archived assembly manifests retain the complete overlap, reimport and analysis options for [6]helicene, α-D-glucopyranose and dihydroxyacetone. All scripts use portable paths.

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